MIL-HDBK-845
31 December 1992

MILITARY HANDBOOK

STANDARD POWER SUPPLY APPLICATIONS HANDBOOK



AMSC N/A FSC 6130

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1. SCOPE

1.1 Introduction. This handbook is written to be an informational aid for electronic system designers who use or plan to use Standard Power Supplies (SPS). Included are performance characteristics of SPS, electrical and mechanical interface requirements, and general applications guidance and instructions.

1.2 Background. The term Standard Power Supply (SPS) refers to any one of a group of low voltage power supplies which are designed and manufactured to meet requirements established under the Navy's Standard Hardware Acquisition and Reliability Program (SHARP). The purpose of this effort is to provide system designers with a group of power supplies with varied standard input/output capabilities which meet established performance, reliability, and maintainability criteria. SPS represents one step toward realizing system acquisition through the use of standardized hardware usage. By selecting from this group, system designers can reduce many of the initial acquisition costs associated with power supply design, test, and implementation. SPS designs are often born out of the need for certain power supply capabilities in new electronic systems. As a result, the first user often influences which capabilities are built into each supply. The standardization philosophy builds on the needs of individual system requirements and outlines a power supply which can be used in multiple applications.

1.3 Categorization. Standard Power Supplies are categorized into two groups: Shipboard Power Supply Products (SPSP) and Airborne Power Supply Products (APSP). SPSP are intended for use specifically in shipboard applications and are designed in accordance with MIL-P-24764. APSP are designed for airborne applications to the requirements of MIL-P-29590.

1.4 Organization. This document addresses several significant applications criteria for SPS. The information is separated between SPSP and APSP for each criteria. In those cases where the information is appropriate to both airborne and shipboard applications, it is not repeated.

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2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation.

SPECIFICATIONS

FEDERAL

QQ-P-35	- Passivation Treatments for Corrosion-Resistant Steel

MILITARY

MIL-P-24764	- Power Supplies, Shipboard, Electronic, General Specification for
MIL-C-28748	- Connector, Plug and Receptacle, Rectangular, Rack and Panel Solder Type and Crimp Type Contacts, General Specification for
MIL-C-28754	- Connectors, Electrical, Modular, and Component Parts, General Specification for
MIL-M-28787	- Modules, Standard Electronic, General Specification for
MIL-P-29590	- Power Supplies, Airborne, Electronic, General Specification for
MIL-C-38999	- Connector, Electrical Circular, Miniature, High Density Quick Disconnect (Bayonet, Threaded and Breech Coupling), Environment Resistance Removable Crimp and Hermetic Solder Contacts, General Specification for
MIL-C-83527	- Connectors, Plug and Receptacle, Electrical, Rectangular Multiple Insert Type, Rack to Panel, Environment Resisting, 150°C Total Continuous Operating Temperature

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MIL-E-85726 - Enclosure, Standard Avionics, Forced Air-Cooled, General Specification for

STANDARDS

MILITARY

MIL-STD-454 - Electronic Equipment, Standard General Requirements for
MIL-STD-461 - Electromagnetic Emission and Susceptibility Requirements for the Control of Electromagnetic Interference
MIL-STD-462 - Electromagnetic Interference Characteristics, Measurement of
MIL-STD-704 - Aircraft Electric Power Characteristics
MIL-STD-1389 - Design Requirements for Standard Electronic Modules
MIL-STD-1399 section 300 - Interface Standard for Shipboard Systems
MIL-STD-2036 - General Requirements for Electronic Equipment Specifications
MIL-STD-2038 - Employing Standard Power Supplies, Requirements for
MIL-STD-5400 - Electronic Equipment, Airborne, General Requirements for

HANDBOOKS

MILITARY

MIL-HDBK-217 - Reliability Prediction of Electronic Equipment
MIL-HDBK-241 - Design Guide for Electromagnetic Interference (EMI) Reduction in Power Supplies

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the Standardization Documents Order Desk, Building 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

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NAVMAT P-4855-1A - Navy Power Supply Reliability

(Application for copies should be addressed to the Commanding Officer, Naval Publications and Forms Center, 5801 Tabor Avenue, Philadelphia, PA 19120-5099. Cite stock number 0518-LP-204-4800.)

2.2 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents listed in the DODISS are the issues of the documents cited in the solicitation.

AMERICAN MATERIAL SPECIFICATIONS (AMS) (Listed with SAE)

SOCIETY OF AUTOMOTIVE ENGINEERS, INC. (SAE)

AMS 5604 - Steel Sheet, Strip, and Plate, Corrosion Resistant 16.5Cr - 4.0Ni - 4.0Cu - 0.30 (Cb + Ta) Solution Heat Treated. Precipitation Hardenable.

(Application for copies should be addressed to the Society of Automotive Engineers, Inc., 400 Commonwealth Drive, Warrendale, PA 15096.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

ASTM A693 - Standard Specification for Precipitation-Hardening Stainless and Heat-Resisting Steel Plate, Sheet, and Strip

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103-1187.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

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3. DEFINITIONS

3.1 Acronyms. The acronyms used in this handbook are defined as follows:

(a) APSP - Airborne Power Supply Product.

(b) SPSP - Shipboard Power Supply Product.

(c) SHARP - Standard Hardware Acquisition and Reliability Program.

(d) SPS - Standard Power Supply.

(e) SEM - Standard Electronic Module.

(f) CEEE - Common Electronic Equipment Enclosure.

(g) CCT - Critical Component Temperature.

(h) TCCT - Transient Critical Component Temperature.

(i) MTBF - Mean Time Between Failures.

3.2 Definitions. The following definitions apply.

3.2.1 Mounting surface. Mounting surface refers to that part of the power supply which serves as a thermal interface and which, when mounted, comes in direct contact with the cabinet cooling plate.

3.2.2 Cooling plate. Cooling plate refers to that part of the system rack or cabinet which comes in contact with the power supply mounting surface. The cooling plate acts as a thermal heatsink for power supplies mounted directly to it.

3.2.3 SPSP-QAA. Shipboard power supply products quality assurance activity. Presently, this function is served by the Naval Surface Warfare Center, Crane Division, Code 6023, Crane, Indiana 47522.

3.2.4 APSP-QAA. Airborne power supply products quality assurance activity. Presently, this function is served by the Naval Air Warfare Center, Aircraft Division, Indianapolis, Code 3060, Indianapolis, IN 46219-2189.

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4. SPSP MECHANICAL CHARACTERISTICS

4.1 Introduction. This section provides mechanical information which may be critical in application and use of SPSP power supplies.

4.2 Family. Shipboard Power Supply Products (SPSP) are divided into two families based on packaging configurations.

4.3 Family S1 packaging. Family S1 power supplies are packaged in plug-in style cases with fixed length and height and varying widths (see the appendix). Each package is interfaced via a recessed connector opening located on the bottom of the supply. The mounting surface area serves as a thermal interface to conduct heat from internal components. The power supply is secured to the cooling plate/connector interface by six (6) hold-down screws around the periphery of the supply and by two (2) connector jackscrews (near the connectors on styles 2, 3, 4, and 5). Proper torquing of these screws (as specified in the associated detail specification) is critical for mechanical interface integrity, electrical connector protection, and adequate heat exchange to the cooling plate. Figure 1 shows style 3 power supplies used in systems designed for water cooled common electronic equipment enclosures (CEEE) as well as systems utilizing air cooled cabinet configurations. The style 3 supply is sized to allow two (2) power supplies to be mounted vertically on each CEEE drawer (figure 2).

4.3.1 Family S1 key code. The functionality of individual family S1 power supplies is distinguished by a key code assignment. Each power supply has four holes located on the bottom surface of the unit as shown in the appendix. The top two holes (closest to the electrical connector) are referred to as stationary keying holes and are located in the same area on all family S1 power supplies. The bottom two holes are referred to as personality keying holes and designate the personality of the power supply. Both sets of holes are interfaced by keying dowel pins which must be provided on the cooling plate. Key coding is used to designate placement of the personality keying pins, for alignment of the unit on the cooling plate, and also to prevent unintentional installation of the wrong power supply. Power supplies with the same key code designation are fully interchangeable.

4.3.2 Keying pins. The cooling plate/connector interface should be equipped with engaging dowel pins to mate with the power supply keying holes. The recommended diameter of the keying pins is 0.250 $\pm$ .0005 inches (maximum allowable diameter of 0.255 inches). Polished drill rod with this dimension and tolerance is readily available. The recommended height of the

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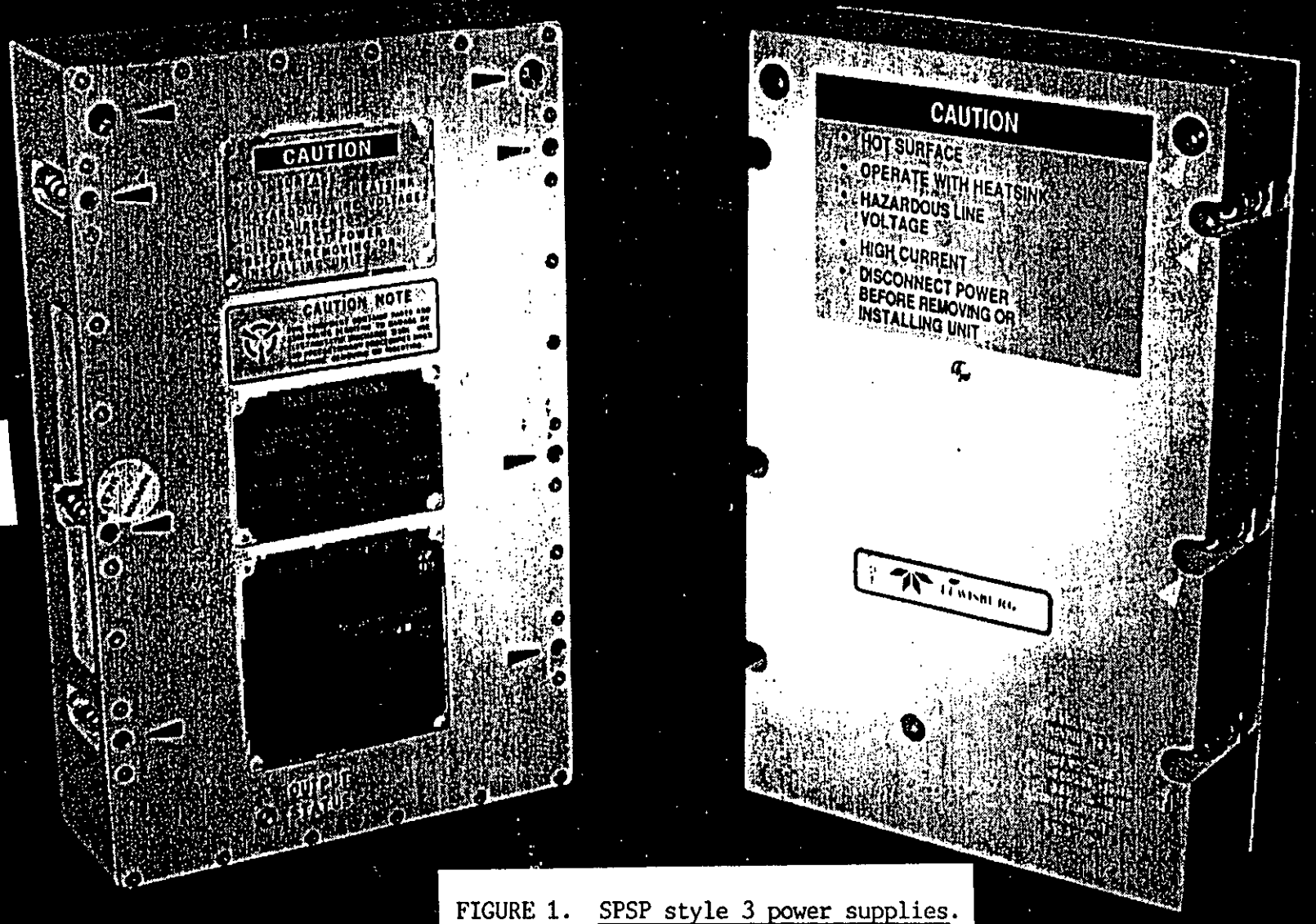


FIGURE 1. SPSP style 3 power supplies.

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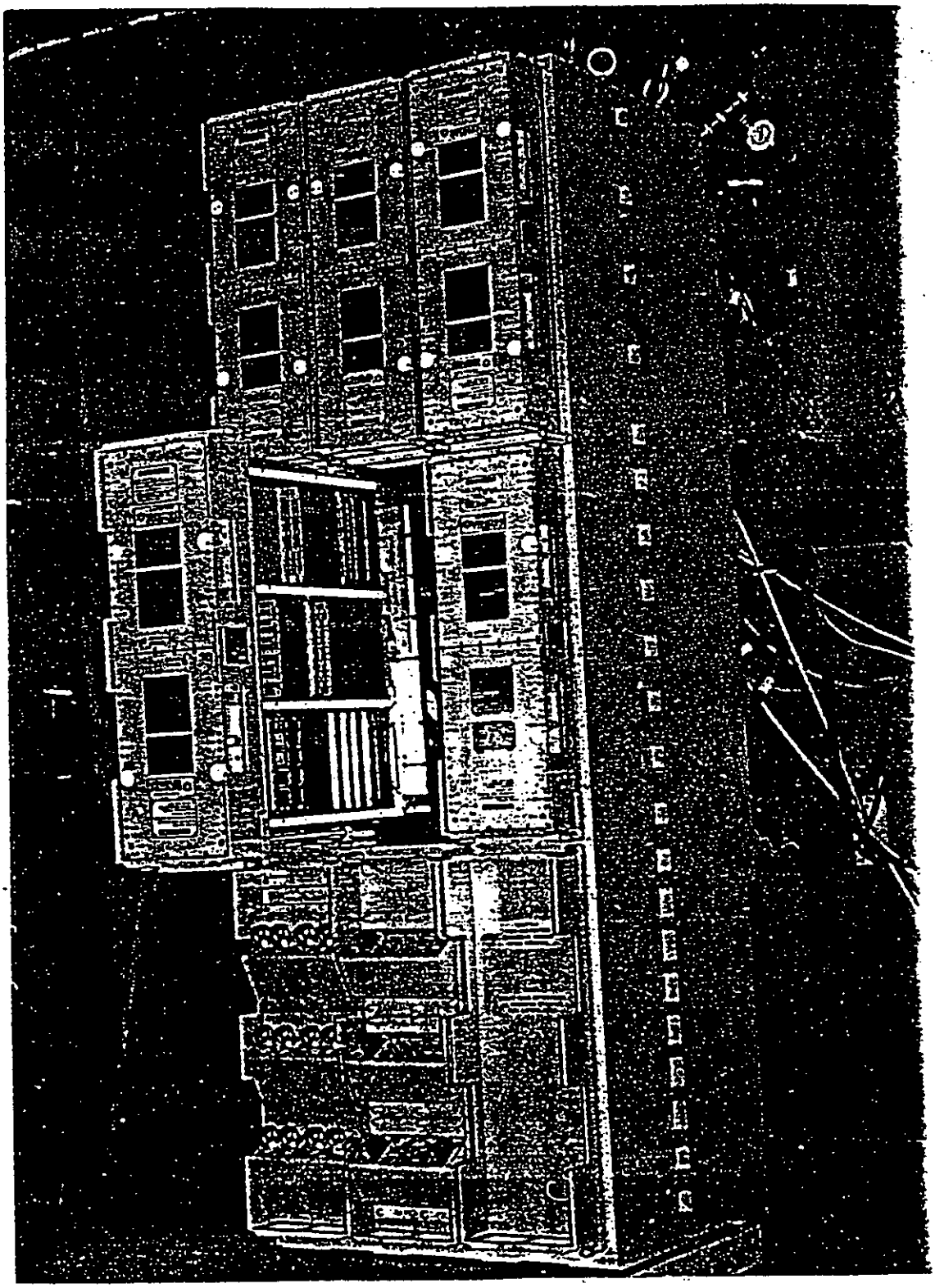


FIGURE 2. CEEE cabinet configuration.

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keying pins is 0.875 ± .060 inches above the cooling plate surface. Under no circumstance should the pin be less than 0.740 inches or greater than 1.000 inches in height. This length pin will ensure two things. First, incompatibly keyed power supplies cannot be engaged to the connector. Also, in the event that the power supply is hung on the cabinet drawer without being fully engaged (via the hold down screws), the keying pins will help support the power supply--taking some of the weight off the electrical connector.

4.3.3 Family S1 connectors. Family S1 power supplies use MIL-C-28748 type connectors. A listing of the connectors used on these power supplies is given in MIL-P-24764. Connectors should be positioned on the cooling plate to mate with the power supply per the dimensions given in the appendix. The connectors should be mounted with a maximum height of 0.680 inches above the cooling plate surface. The minimum mounting height must ensure that the connector contacts and sockets have a 0.062 inch minimum wiping engagement.

4.3.3.1 Connector brackets. A connector bracket design which supports the mating connectors on floating mounts should be considered for style 3 power supplies. A floating mount allows for alignment during power supply engagement. Bracket dimensions are critical for proper interfacing with the power supply. The maximum bracket width should not exceed 0.965 inches. The SPSP-QAA has information on connector bracket drawings and design.

4.3.3.1.1 Floating connectors. The term "floating connector" refers to the method of aligning the electrical connectors (on the connector bracket) within the allowable power supply connector location tolerance to allow proper interfacing of each power supply to the cooling plate and connector bracket. Once aligned, each connector is then firmly secured to the connector bracket. Various jigs and tools have been developed to aid in the alignment process. Information can be obtained from the SPSP-QAA.

4.3.3.2 Connector mating contacts. Connector keying/guide pins and receptacles, used for aligning the connector during power supply installation, should be provided per MIL-C-28748.

4.3.3.3 Connector engagement. Proper power supply installation and removal in accordance with the instructions on the power supply is necessary. Care should be exercised when engaging power supplies to the connector interface to avoid connector damage due to forced misalignment. Once aligned properly, connector jackscrews should be tightened alternately one full turn at a time until snug and then torqued to the specified 32 ± 3 in-lbs.

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4.3.4 Cooling plate characteristics. The cooling plate surface should be flat to within 0.0005 inch-per-inch with an overall flatness (over the complete area to be mated with the power supply) not to exceed 0.005 inches. A suitable cooling plate material (used in the CEEE cabinet) is stainless steel plate 17-4 PH, Cond. A, surface finish per AMS 5604 or ASTM A693, passivated surface per QQ-P-35 type I, II, or III.

4.3.4.1 Cooling plate engagement. Captive hold down screws should be tightened alternately until snug and then torqued to the specified 32 ± 3 in-lbs. Hold down screws are 0.190 inch-32 UNF-3A type socket head screws. It is recommended that hole locations on the cooling plate for the hold down screws be threaded (or fitted with threaded inserts) to a minimum depth of 0.40 inches. The connector jackscrews (see 4.3.3.3) must be secured first (prior to hold down screws) during power supply installation and released last during power supply removal.

4.3.4.2 Sealing devices. Family S1 power supplies are equipped with a mounting surface sealing device to prevent moisture from wicking into the thermal interface area between the power supply mounting surface and cooling plate or from entering the cabinet through the cable opening in the cooling plate. There is no need to apply any other sealant material to the interface area as long as the cooling plate has the characteristics of 4.3.4. The cooling plate must not have recesses, counter sunk screws, grooves, etc. in the area corresponding to the allowed area for the sealing device.

4.3.5 Thermal considerations. Conversion of input power to regulated output DC power cannot be accomplished at 100 percent efficiency. Inefficiencies in power supply operation are manifest in the form of heat. Heat is one of the major contributors of failure in electronic devices. As a result, adequate thermal management must be maintained to conduct heat away from the power supply and to assure proper operation.

4.3.5.1 Mounting surface temperature. The maximum operating power supply mounting surface temperature is not to exceed 60°C during operation of the supply (power supplies are specified to operate for 1 hour at a transient thermal operating temperature of 80°C). The system designer should ensure compliance to this requirement under worst-case environmental and operating conditions.

4.3.5.2 Heat dissipation. Proper operation of the power supply can be assured by providing adequate heatsinking through the cooling plate interface. Switch-mode power supplies operating at nominal efficiencies can still generate large amounts of heat. The AA3A power supply, for example, can provide

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up to 812 Watts of output power. Operating at the minimum allowed efficiency of 70 percent, this power supply can dissipate up to 348 Watts of power in the form of heat through the mounting surface area.

4.3.5.3 Thermal resistance. The thermal resistance between the power supply mounting surface and the cooling plate surface should be minimized to allow high rates of heat removal. The torquing requirement of 32 inch-lbs for each of the six (6) mounting screws combines to create 65 lbs per square inch (psi) interface pressure. Analysis has shown that the 65 psi pressure, along with the cooling plate flatness and finish requirements, provides for a thermal resistance to maintain a 7.5°C interface temperature difference (called out in MIL-P-24764). Therefore, the cooling plate temperature must be maintained at least 7.5°C lower than the specified maximum operating temperature (52.5°C for family S1 power supplies). Failure to comply with the torque or mating surface requirements may increase thermal resistances to a level where the specified interface temperature difference cannot be maintained. Whatever the temperature difference is determined to be (through analysis or measurement), the cooling plate temperature must be regulated to maintain a maximum of 60°C at the power supply mounting surface.

4.3.5.4 Cooling medium. SPSP users can choose a cooling scheme best suited for each application provided the required mounting surface temperature is maintained. The power supply must be mounted to a heat exchanger. The heat exchanger can have various configurations and utilize different heat transfer media. Forced air and forced water cooling schemes, which remove heat which has been conducted from the power supply to the cooling plate, are used extensively. A forced water approach is used in the CEEE cabinet (see figure 3). Lower power SPS may be adequately cooled by natural convection provided good thermal design practices are utilized.

4.3.5.5 Thermal grease. Thermal grease (and similar compounds) should be avoided for a number of reasons.

(a) the thickness and application of thermal grease compounds is not easily controllable

(b) excessive grease tends to migrate to other (undesirable) locations

(c) thermal grease compounds tend to trap and retain foreign material--oftentimes decreasing thermal conductivity

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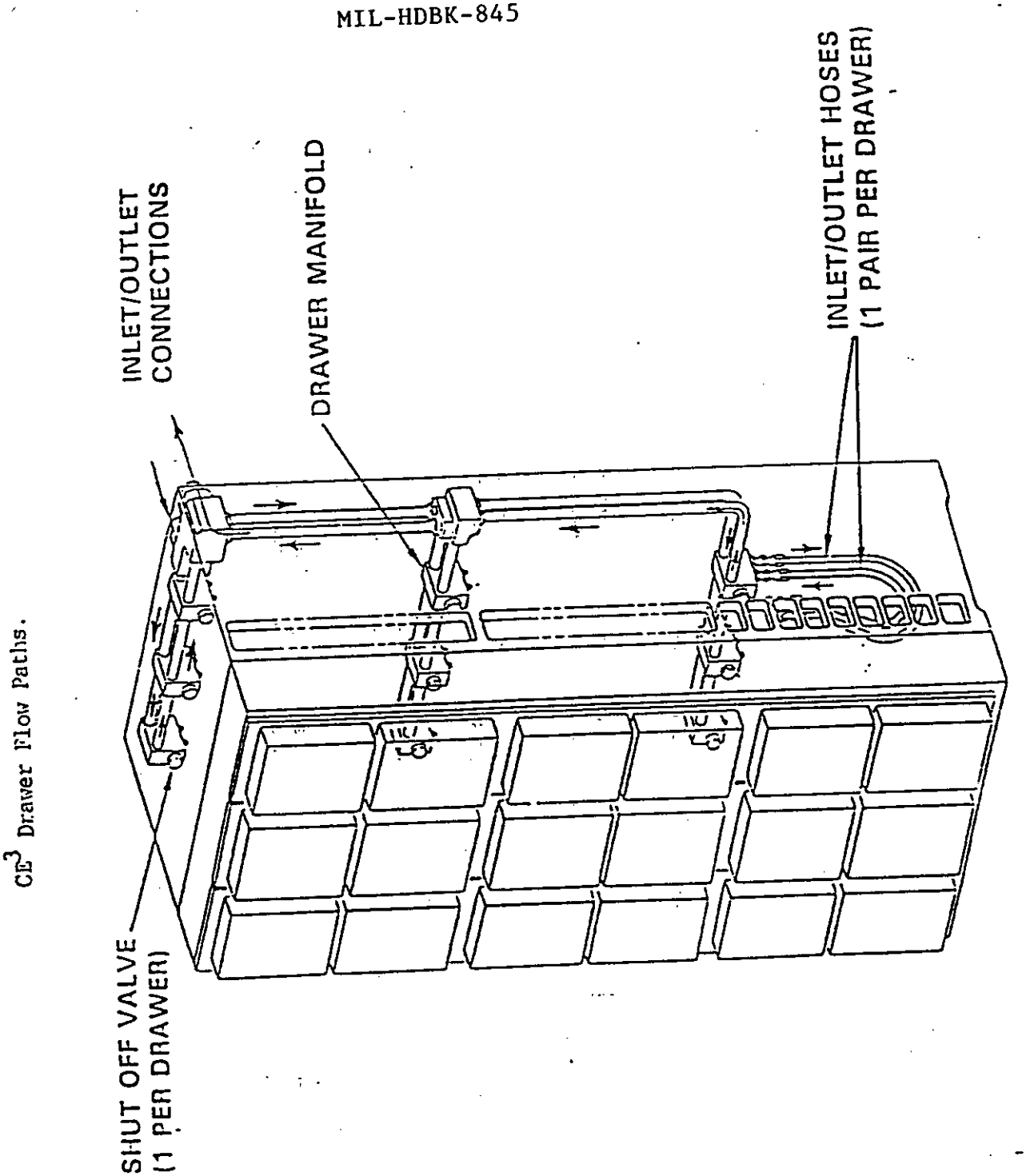


FIGURE 3. Forced water cooling in the CEEE cabinet.

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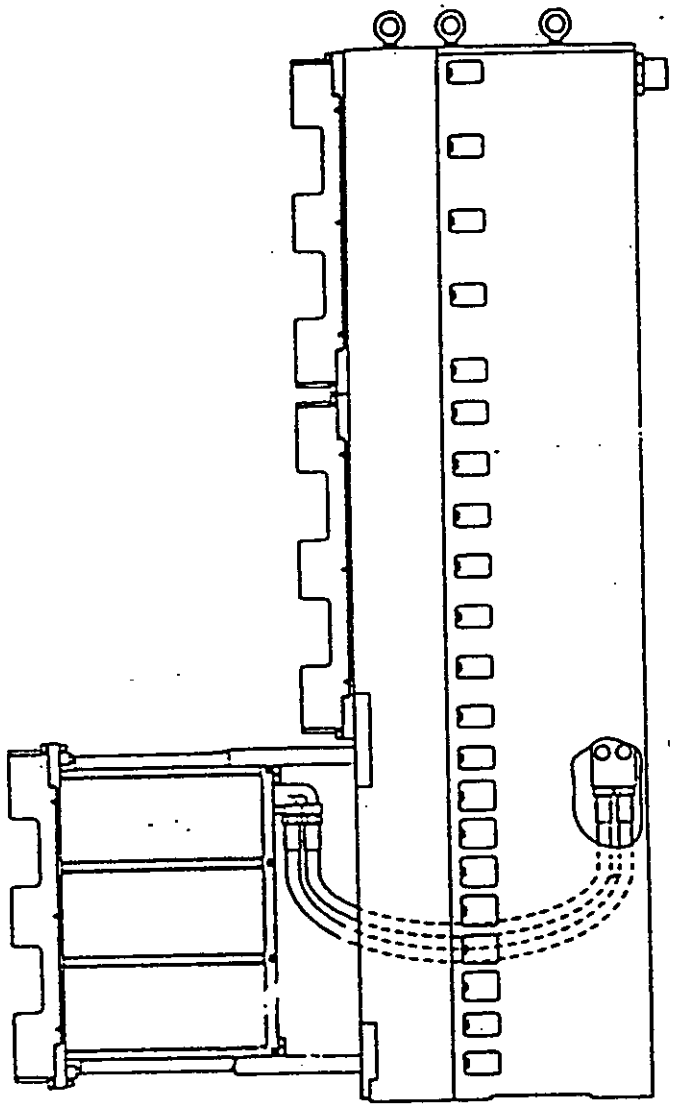


FIGURE 3. Forced water cooling in the CEEE cabinet - Continued.

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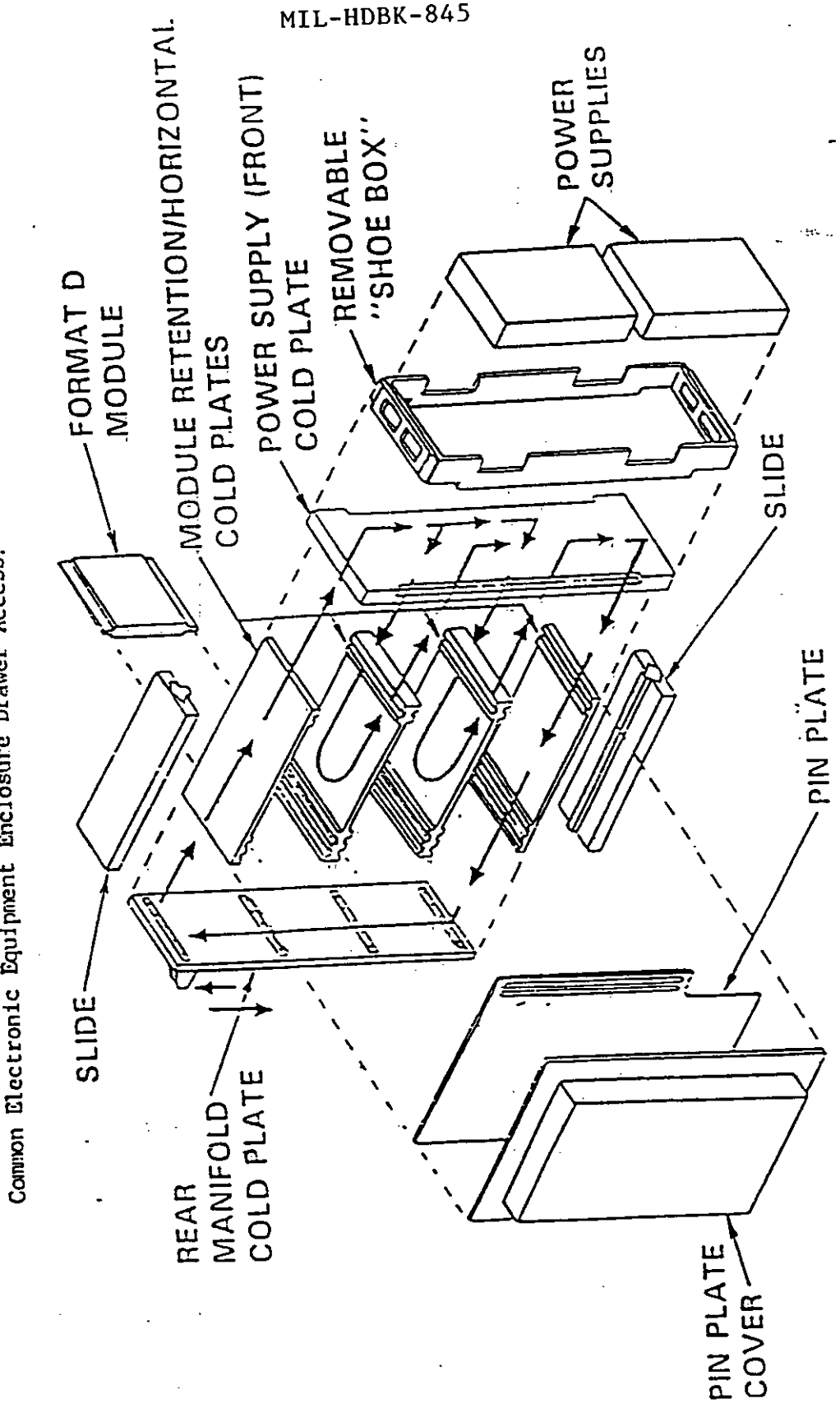


FIGURE 3. Forced water cooling in the CEEE cabinet - Continued.

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(d) replacement of thermal grease following removal of a power supply is not guaranteed

Power supplies are specified to meet performance requirements without utilizing thermal grease on the power supply mounting surface.

4.4 Family S2 packaging. Family S2 power supplies are packaged as Standard Electronic Module (SEM) compatible units per MIL-STD-1389 in five (5) formats as shown on figure 4. The module thickness is allowed to vary in discrete increments per MIL-P-24764.

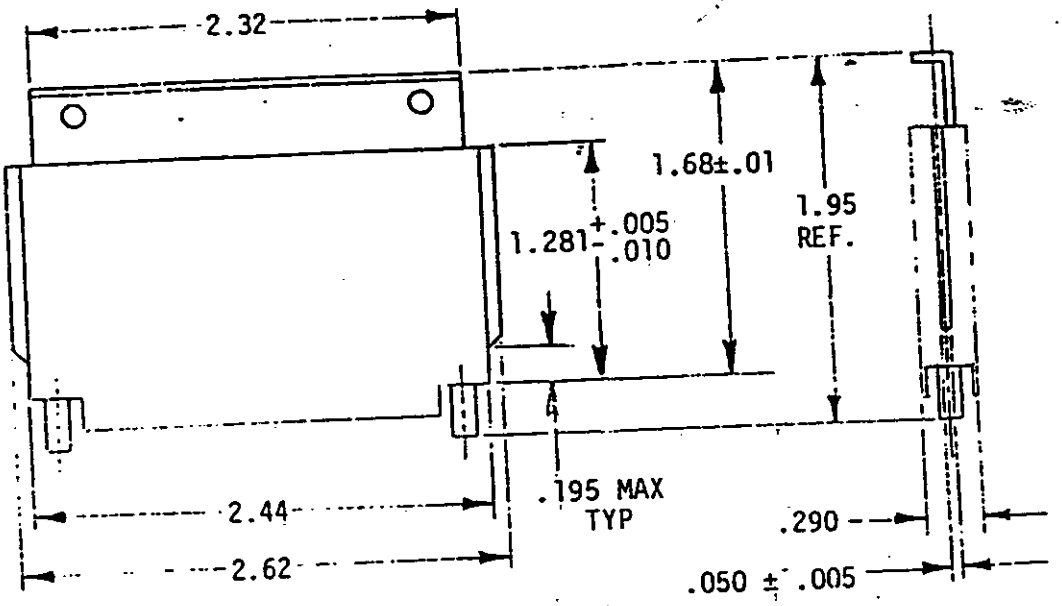
4.4.1 Family S2 key code. Keying pins are located on SEM units on each end of the connector interface area. Keying pins are designated as Alpha and Beta pins. Alpha pins are located adjacent to the lowest numbered connector contact and beta pins are located adjacent to the highest numbered connector contact. Keying pin orientations are defined by the key code and as specified in MIL-P-24764. Appropriately oriented mating receptacles must be installed as a part of the mating connector. Power supplies with the same key code designation are fully interchangeable.

4.4.2 Family S2 connectors. Family S2 power supplies use blade type connectors per MIL-C-28754. The number of contacts may vary with power supply size format and individual design. Information on connectors for individual units can be found in the detail specification or in appendices A through E of MIL-STD-1389.

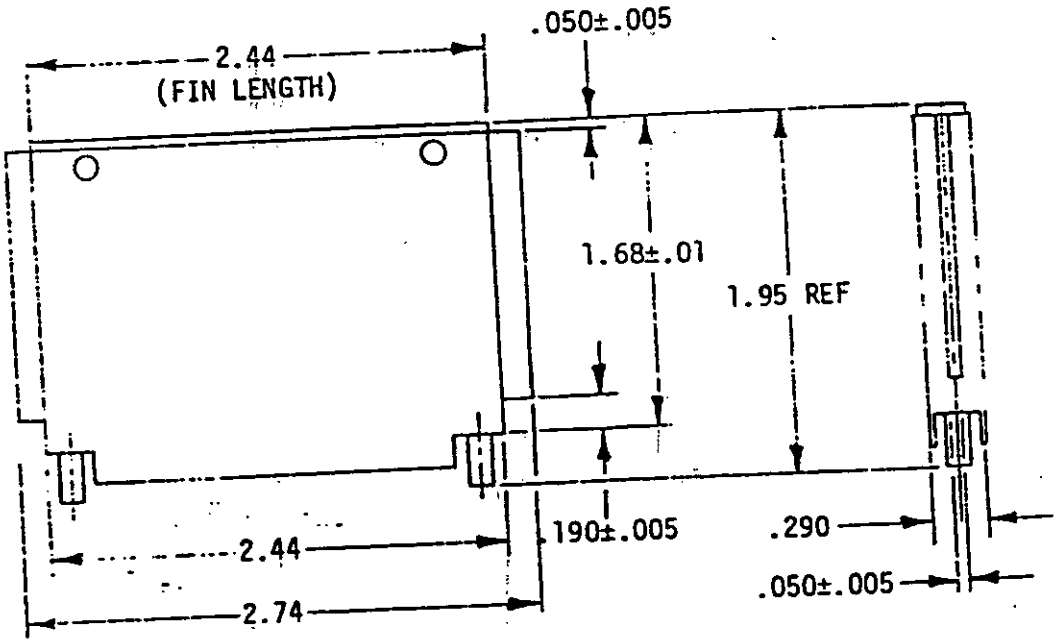
4.4.3 Thermal considerations. Family S2 power supplies may have more than one thermal interface. Guide ribs are required on SEM power supplies for installation alignment and heat transfer. These guide ribs must be clamped or otherwise secured adequately in the card rack to assure good heat conduction from the supply. For some module configurations, additional heat removal may be needed through a baseplate mounting surface area. In multi-ribbed modules, due to mechanical tolerances, only one set of guide ribs can be relied on for major heat transfer when clamping to a card rack mount. The detail specification and/or appendices A through E of MIL-STD-1389 should be referenced for individual power supply thermal requirements.

4.4.3.1 Thermal resistance. Thermal resistances between card racks/cooling plates to family S2 thermal interfaces should be minimized. The cooling plate temperature must be regulated to maintain a maximum power supply interface temperature of 60°C. This maximum temperature must not be exceeded under worst-case operating and environmental conditions.

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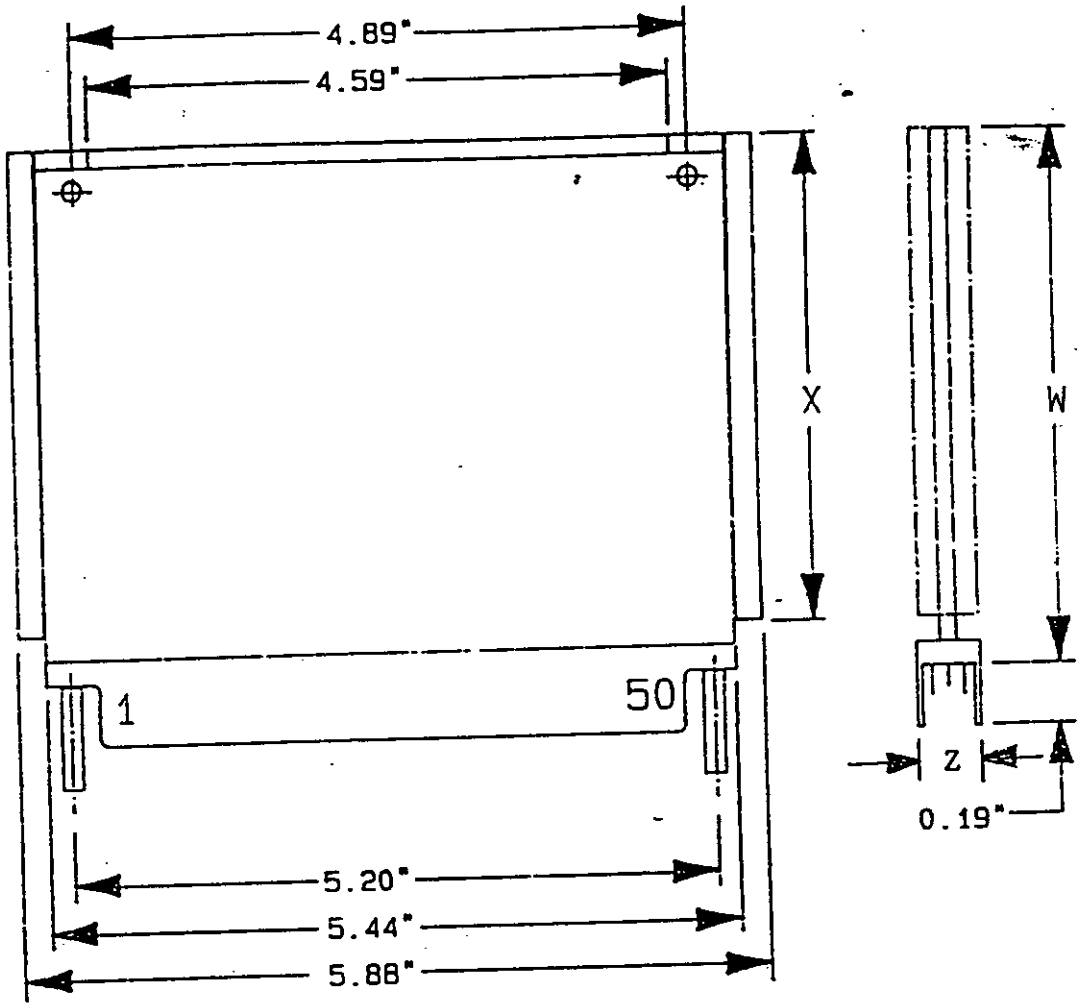
Format A



Format B

FIGURE 4. Family S2 SEM formats.

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FORMAT	X	W	Z
C	3.47"	3.79"	0.28"
D	4.00"	4.56"	0.28"
E	5.86"	6.41"	0.38"

FIGURE 4. Family S2 SEM formats - Continued.

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5. APSP MECHANICAL CHARACTERISTICS

5.1 Introduction. This section provides mechanical information applicable to APSP power supplies. Where this information is similar to SPSP, reference is given.

5.2 Family. Airborne Power Supply Products (APSP) are divided into two families based on packaging configurations.

5.3 Family A1 packaging. Family A1 power supplies are packaged as Weapon Replaceable Assembly (WRA) compatible units in accordance with MIL-E-85726. MIL-P-29590 and associated detail specifications should be referenced for individual configurations and dimensions.

5.3.1 Family A1 key code. Family A1 power supplies are keyed in accordance with MIL-C-38999 or MIL-C-83527 and the associated detail specification.

5.3.2 Family A1 connectors. MIL-P-29590 and associated detail specifications should be referenced for family A1 connector types.

5.3.3 Thermal considerations. Family A1 APSP power supplies are specified to operate over a temperature range of -54 to 71°C in accordance with the MIL-STD-5400, Class 2X environment. These supplies are intended to be cooled using forced air cooling schemes. The system designer must ensure that adequate cooling air is available to accommodate thermal management of family A1 supplies. Reference should be made to the associated detail specification.

5.4 Family A2 packaging. Family A2 power supplies are packaged as SEM compatible units in formats B, C, and E configurations (see 4.4).

5.4.1 Family A2 key code. See 4.4.1.

5.4.2 Family A2 connectors. Some special considerations must be made for airborne power supplies concerning altitude and corona. These considerations may dictate spacing and arrangement of connector contacts and locations. Reference should be made to the associated detail specifications.

5.4.3 Thermal considerations. Family A2 APSP are specified to operate over a temperature range of -55 to 85°C. In addition, power supplies are specified to operate at a transient temperature of 105°C for a minimum of 1 hour under full load conditions.

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6. SPSP ELECTRICAL

6.1 Introduction. This section includes electrical information concerning the application and use of SPSP power supplies.

6.2 Input power. Shipboard Power Supply Products (SPSP) are designed to operate in a shipboard environment where the standard service supplied by the ship's alternators is 440 VAC, 60 Hz, 3-phase, delta-connected, ungrounded power designated Type I power in MIL-STD-1399, section 300. The specified line-to-line input voltage for SPSP is 115 Vrms--obtained from step-down transformers on the shipboard system.

6.2.1 155 VDC input. SPSP are also specified to operate from 155 VDC power applied between any two (2) of the power supply's three-phase inputs. In systems requiring multiple power supplies, the designer should always consider using 155VDC as the prime power source to the system. Due to the ungrounded nature of shipboard power, harmonic line currents must be controlled. MIL-STD-1399, section 300 specifies a 3 percent harmonic current limitation for all shipboard systems rated at 1kVA or more. Power supply rectification of the Type I AC service creates harmonics which exceed this limit requiring line conditioning electronics control or extensive filtering. These low frequency harmonics are not created by the power supply when DC is used as the prime power source. Systems which use multiple power supplies may gain benefits from rectifying the AC source to DC and filtering at a central location and then providing the DC input to the power supplies. The DC input capability of SPSPs can be advantageous in platforms which have only 400Hz service. The 400Hz service can be centrally rectified/filtered and then provided as 155VDC input.

6.2.1.1 Obtaining 155 VDC power. Several methods are available for converting ship's AC power to 155 VDC and controlling the harmonic current to less than 3 percent. One of the most practiced methods is with a polyphase transformer, diode bridge, and LC filter. This method is described as the rectifier conditioner portion of the Navy Standard Electronic Power System in MIL-STD-2036.

6.2.2 DC auxiliary inputs. The DC auxiliary inputs are included on some SPSP to provide a location for back-up or uninterruptible power provisioning.

6.2.3 Input current considerations. Switch-mode power supplies exhibit high inrush current characteristics at turn-on due to the rectifier and large capacitor input circuitry. As a result, system designers must make adequate considerations in

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breaker/fuse coordination. SPS are specified to a maximum inrush current limit of 10 times the steady state true RMS or DC value. An AA3A power supply, for example, operating with DC input under full load conditions at an efficiency of 70 percent will normally require 7.0 amps input current. At startup, a 70 amp current peak may exist. The duration of the inrush current may be as long as 500 usec. As a result, adequate breaker/fuse sizing is mandated to prevent false tripping/blowing under startup conditions.

6.2.4 Low input voltage conditions. Since switch-mode type power supplies are constant power devices (see 8.4.1), they tend to draw more input current as the input voltage level decreases. SPSPs are specified to limit input current to two (2) times the input steady state true RMS or DC current at maximum rated load and nominal input voltage. This 2 times input current limit applies at any steady state input voltage from 0 volts to the maximum specified input voltage (and at turn-on after the inrush current has occurred). This means that unless otherwise limited, the power supplies could possibly furnish full load current with as low as 50 percent of the input voltage. This operating range, even though not guaranteed, must be considered in regard to power distribution design and stability.

6.2.5 Emergency input conditions. SPSP supplies are also specified to operate under emergency input voltage excursions up to 135 percent of the nominal input value. This would be a maximum AC voltage of 155 VRMS or 210 VDC.

6.2.6 Spike voltage. The spike voltage requirement specifies the capability of the power supply to withstand, without damage, a high voltage, short duration voltage spike on the input in accordance with MIL-STD-1399, section 300. Most power supplies will not withstand the full 1,000 volt spike amplitude specified for the 115 VAC power line. As a result, the user equipment will have to clip the voltage spike, somewhere external to the power supply input, to the level specified in the associated detail specification.

6.3 Control functions. External control functions available on SPSP may include: remote ON/OFF capability, output voltage level programming, output overvoltage shutdown control, and synchronization of internal switching frequencies. Refer to the power supply detail specification to determine which functions are available on each power supply design configuration.

6.3.1 Remote ON/OFF. The remote ON/OFF function serves to turn the power supply output on and off without interruption of the input power service. The remote switch interface is

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compatible with both transistor-transistor logic (TTL) level signals (0.0 - 5.5 Vdc) and direct wiring control. A TTL logic high (2.0 Vdc min., 100 uA max.) or a remote switch interface open circuit will turn the power supply output OFF. A TTL logic low (0.8 Vdc max., 3.2 mA min.) or a short circuit between the remote switch and the remote switch return will turn the power supply on. Power supplies with remote ON/OFF capability will not provide output voltage unless the remote switch interface is configured for the ON condition. Any power supply latch condition, such as an overvoltage latch, may not be reset by cycling the remote ON/OFF. This reset may only be accomplished by cycling the input power.

6.3.2 Voltage programming. The voltage programming interface is used to select output voltage levels in discrete levels via pin programming as specified in the power supply detail specification. For those power supplies with two output voltage levels, an open circuit at the voltage programming interface programs the supply to the lower output voltage and a closed circuit (voltage program contact to voltage program return contact) programs the supply to the higher level. See the detail specifications for configuration instructions for those supplies which have more than two output voltage levels. The voltage programming control interface is not TTL (or other logic level) compatible. It is also not necessarily isolated from internal voltage control circuitry. Application of analog signals to the voltage programming interface may result in an adverse effect on the power supply output voltage.

6.3.3 Synchronization. Power supplies may be specified to have a control interface which allows the power supply internal switching frequency to be synchronized with the switching frequency of another power supply or other external source. The power supply detail specification should be referenced to determine if synchronization capability is provided.

6.4 Output. Each SPSP key code has output capabilities unique to that supply. In addition, several protection features may be included to prevent external damage in the event of power supply failure.

6.4.1 Output voltage sensing. Power supplies may be equipped with output voltage sensing capability. This function causes the output voltage to be regulated to the specified level at the location where the sense leads are connected. A maximum length at which remote sensing can be accomplished is specified in the detail specification for those power supplies which have this capability. The user should be aware that power supply operation may be adversely affected when remote sensing is being used and there is significant inductance in the load leads.

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6.4.2 Overcurrent protection. Each SPSP is equipped with overcurrent protection circuitry to prevent damage due to output short circuiting or overloading. Overcurrent protection is initiated in the range of 100 percent to 125 percent of full rated load. Two methods of implementing overcurrent protection exist in SPSP designs. Some designs incorporate "fold-back" circuitry as a protection scheme. When overcurrent condition is reached, the output voltage drops off proportionately as the current increases. As a result, a maximum operating current level is controlled. The second scheme uses a "hiccup" mode protection. When an overcurrent condition is detected, the output shuts off. The output then turns on intermittenly and checks to see if the overcurrent condition has been cleared. Regardless of the method, it is required that normal operation be resumed when the overcurrent condition is removed.

6.4.3 Overvoltage protection. Each SPSP is equipped with output overvoltage protection. This protection is initiated when output voltage exceeds rated level by 15 percent or 1.2 Vdc -- whichever is greater. Upon an overvoltage, some power supply designs may latch off while others do not (see the detail specification). It may be necessary to take the input voltage to zero in order to reset the output protection circuitry. The remote ON/OFF interface cannot be used to reset the output protection circuitry.

6.4.4 Hold-up time. Hold-up time refers to a power supply's ability to provide regulated output for a period of time after input power to the supply has been removed. SPSPs are specified to have a 50 usec hold-up time. When 155 VDC is used as the primary power to the SPSP, hold-up time can be increased by adding hold-up circuitry (generally, added capacitance) across the input lines (see MIL-STD-2036).

6.5 Status signals. SPSP may be equipped with status signals which indicate various operating parameters of the supply. The power supply detail specification should be referenced to determine which status signals are included on each supply.

6.5.1 Overtemperature indication. SPSP supplies are required to contain two (2) overtemperature indication interfaces (see figure 5). Both of these interfaces are thermally activated, normally open switches which close when the overtemperature condition is satisfied.

6.5.1.1 Unsafe temperature indication. The unsafe temperature interface is nominally set to close 10°C above the worst case maximum component temperature evident with a mounting surface temperature of 80°C (TCCT). Unsafe temperature provides

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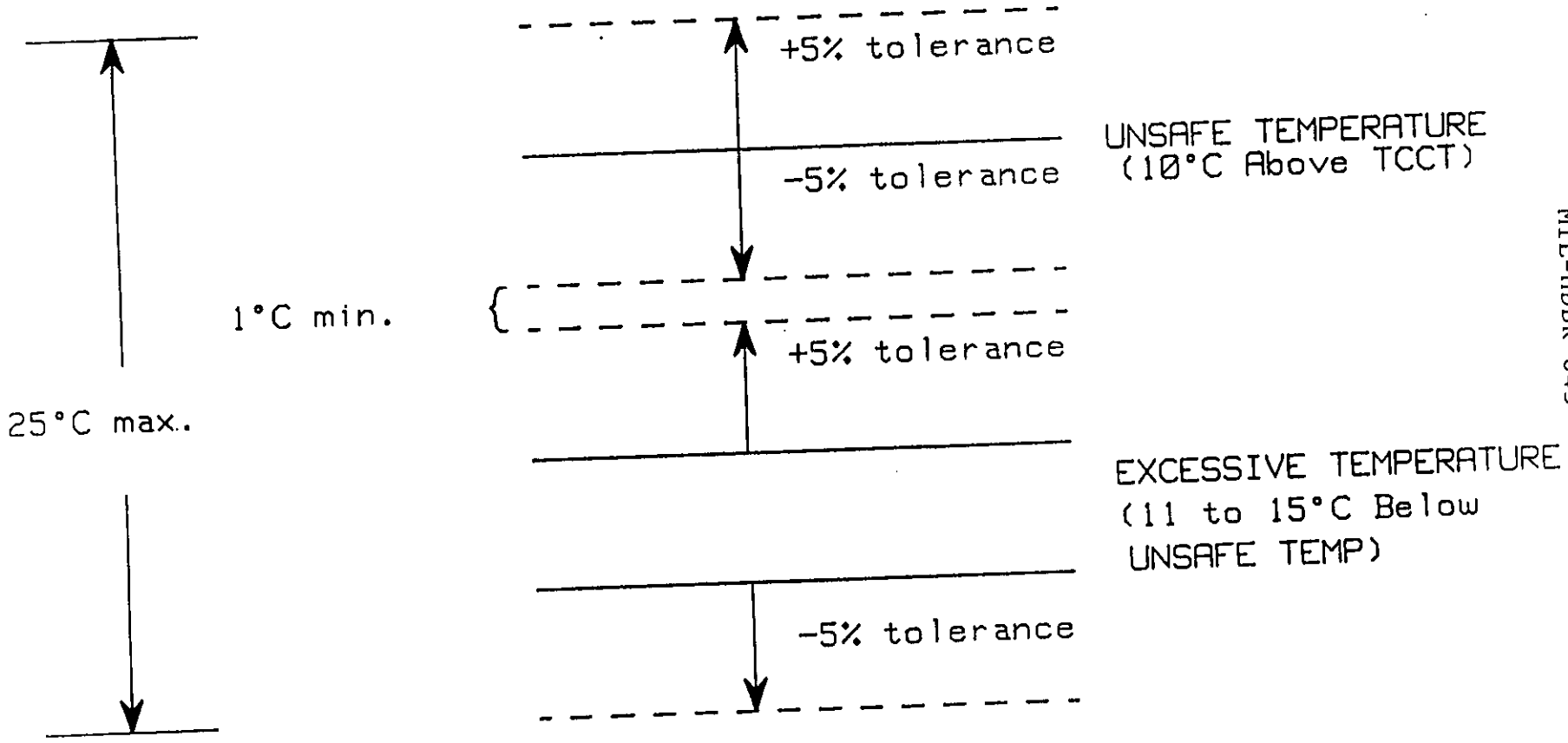


FIGURE 5. Overtemperature settings.

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indication that internal components are operating at or near their maximum thermal limit. Power supplies operated with an unsafe temperature indication are subject to damage or failure.

6.5.1.2 Excessive temperature indication. The excessive temperature interface is nominally set 11 to 15°C lower than the unsafe indication. Excessive temperature indication is given to allow attention to the unit before unsafe indication is given.

6.5.2 Power interrupt. Power supplies may be specified to provide a power interrupt signal. This function monitors internal circuitry and provides a low level (0.4 VDC maximum, 5.0 mA, minimum) signal to indicate that normal power is being applied. If input power is interrupted, the power interrupt signal provides a high level signal or high impedance "off" state indication.

6.5.3 Output status. The output status function acts as a monitor for each output voltage on the unit. Status will be indicated in two ways: (1) a green LED located on the power supply cover, (2) a status signal interface at the electrical connector. The "Go" condition is indicated by a lighted LED or by a low (less than 0.4 Vdc) status signal. The status signal logic is specified such that an out of tolerance condition is indicated by an "off" light or a logic high or high impedance state. This way any power supply out of tolerance condition, including total loss of input or output power, will give a fault indication.

6.5.4 Malfunction. The malfunction signal is an indicator of internal fault condition or interruption of input power. Malfunction status is indicated in two ways: (1) a green LED located on the power supply cover, and (2) a status signal interface at the electrical connector. Normal operation is indicated by a lighted LED or by a low level (less than 0.4 VDC) malfunction signal. High level signal or LED off state occurs when one of the conditions in table I is encountered. It should be noted that interruption of input power will prompt the LED off and may not be indicative of internal, power supply related, malfunction. The malfunction signal is intended to allow fault isolation to the faulty power upply in a paralleled configuration.

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TABLE I. Malfunction combinations. 1/

Output malfunction (MF)	Output under-voltage (A)	Output over-current (B)	Allowed input voltage (C)	On/off remote switch ON (D)	Other internal faults (E)	Output over-voltage (F)
1	Yes	No	Yes	Yes	-	-
2	-	-	Yes	-	Yes	-
3	-	-	-	-	-	Yes

1/ Dash denotes no requirement. $MF = (A\ \overline{B}\ C\ D) + (C\ E) + F$

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7. APSP ELECTRICAL

7.1 Introduction. This section contains electrical information concerning the application and use of APSP power supplies.

7.2 Input power. Airborne Power Supply Products (APSP) are designed for airborne operation where the standard power service is defined by MIL-STD-704. The standard power services defined are single or three phase, 115 VAC line to neutral, 400 hertz (Hz), 28 VDC or 270 VDC.

7.2.1 Family A1 input power. Family A1 power supplies can be specified to operate from any of the MIL-STD-704 power sources.

7.2.1.1 Auxiliary inputs. Family A1 power supplies may be equipped with auxiliary inputs. A power supply specified to operate from AC may also be required to operate with an auxiliary 28 VDC or 270 VDC input.

7.2.2 Family A2 input power. MIL-STD-704 defines the electrical power characteristics at the input terminals to the electronics enclosure. A typical implementation using family A2 power supplies will have an EMI filter and distribution conductors between the enclosure input terminals and the A2 power supply. The EMI filters and distribution conductors will introduce some voltage drop between the MIL-STD-704 interface and the input terminals on the A2 power supplies. The family A2 power supplies are designed to include these line losses for compatible operation with MIL-STD-704. The detail specification contains the input operating voltage ranges.

7.2.2.1 270 VDC input. The 270 VDC characteristics for the family A2 power supplies are derived from full wave rectified, three phase, 115 VAC, 400 Hz power. MIL-STD-704 defines characterisitcs for 270 VDC aircraft electric power service. Full wave rectification of the three phase AC voltage results in 270 VDC with characteristics that are similar to, but not identical to, the 270 VDC characteristics defined in MIL-STD-704. A power supply designed to operate with 270 VDC from full wave rectified AC is compatible with the 270 VDC characteristics defined in MIL-STD-704; however, a power supply designed to operate from 270 VDC defined in MIL-STD-704 is not necessarily compatible with 270 VDC derived from full wave rectified AC. The family A2 input power specifications for 270 VDC allow use on either 270 VDC systems as well as the more common three phase AC systems. In either case, voltage losses between the equipment input terminals and the power supply input terminals, due to EMI filtering, power conditioning, and line losses, must be

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considered when designing the system.

7.2.2.2 28 VDC input. APSP includes a line of family A2 power supply modules that operate from 28 VDC where the 28 VDC is defined by MIL-STD-704. Again voltage losses between the equipment input terminals and the power supply input terminals, due to EMI filtering and line losses, must be considered when designing the system.

7.2.3 Input current considerations. Switch-mode power supplies exhibit high inrush current characteristics at turn-on due to the rectifier and large capacitor input circuitry. As a result, system designers must make adequate considerations in breaker/fuse coordination. APSP family A1 are specified to a maximum inrush current limit of ten (10) times the steady state true RMS or DC value. Family A2 are allowed a higher inrush current due to packaging limitations. The inrush current is limited by the product of current squated and time according to the equation:

$$0.2 \leq I^2 t \text{ (amperes squared seconds)}$$

with an initial peak limit ranging from 65 to 100 amperes. Family A2 implementations may require external current limiting circuitry. Adequate breaker/fuse sizing is needed to prevent false tripping/blowing under start-up conditions.

7.2.4 Low input voltage conditions. Since switch-mode type power supplies are contant power devices (see 8.4.1), they tend to draw more input current as the input voltage level decreases. APSP are specified to limit input current ot 1.8 times the input steady state true RMS or DC current at maximum rated load and nominal input voltage. This 1.8 times input current limit applies at any steady state input voltage from abnormal low-line steady state to abnormal high-line steady state input voltages as defined in MIL-STD-704. This requirement also applies at turn-on after the inrush current has occurred. This operation range must be considered in regard to power distribution design and stability.

7.2.5 Emergency input conditions. APSP supplies are also specified to operate under emergency input voltage conditions. MIL-STD-704 defines the emergency characteristics to be the same as for normal operation.

7.3 Control functions. External control functions available on APSP may include: remote ON/OFF capability, output voltage level programming, output overvoltage shutdown control, and synchronization of internal switching frequencies. Refer to the power supply detail specification to determine which

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functions are available on each power supply design configuration.

7.3.1 Remote ON/OFF. The remote ON/OFF function serves to turn the power supply output on and off without interruption of the input power service. The remote switch interface is compatible with both TTL level signals and direct wiring control. A TTL logic high (2.0 VDC min., 100 uA max.) or a remote switch interface open circuit will turn the power supply output off. A TTL logic low (0.8 VDC max., 3.2 mA min.) or a short circuit between the remote switch and the remote switch return will turn the power supply on. Power supplies with remote ON/OFF capability will not provide output voltage unless the remote switch interface is configured for the on condition. Any power supply latch condition, such as an overvoltage latch, may not be reset by cycling the remote ON/OFF. This reset may only be accomplished by cycling the input power.

7.3.2 Voltage programming. The voltage programming interface is used to select output voltage levels in discrete levels via pin programming as specified in the power supply detail specification. For those power supplies with two output voltage levels, an open circuit at the voltage programming interface programs the supply to the lower output voltage and a closed circuit (voltage program contact to voltage program return contact) programs the supply to the higher level. See the detail specifications for configuration instructions for those supplies which have more than two output voltage levels. The voltage programming control interface is not TTL (or other logic level) compatible. It is also not necessarily isolated from internal voltage control circuitry. Application of analog signals to the voltage programming interface may result in an adverse effect on the power supply output voltage.

7.3.3 Synchronization. Power supplies may be specified to have a control interface which allows the power supply internal switching frequency to be synchronized with the switching frequency of another power supply or other external source. The power supply detail specification should be referenced to determine if synchronization capability is provided.

7.4 Output. Each APSP key code has output capabilities unique to that supply. In addition, several protection features may be included to prevent external damage in the event of power supply failure.

7.4.1 Output voltage sensing. Power supplies may be equipped with output voltage sensing capability. This function causes the output voltage to be regulated to the specified level at the location where the sense leads are connected. A maximum

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length at which remote sensing can be accomplished is specified in the detail specification for those power supplies which have this capability. The user should be aware that power supply operation may be adversely affected when remote sensing is being used and there is significant inductance in the load leads.

7.4.2 Overcurrent protection. Each APSP is equipped with overcurrent protection circuitry to prevent damage due to output short circuiting or overloading. Overcurrent protection is initiated in the range of 100 percent to 125 percent of full rated load. Two methods of implementing overcurrent protection exist in APSP designs. Some designs incorporate "fold-back" circuitry as a protection scheme. When overcurrent condition is reached, the output voltage drops off proportionately as the current increases. As a result, a maximum operating current level is controlled. The second scheme limits the output current to less than 125 percent of full rated load while attempting to maintain output voltage.

7.4.3 Overvoltage protection. Each APSP is equipped with output overvoltage protection. This protection is initiated when output voltage exceeds rated level by 15 percent or 1.2 VDC--whichever is greater. Upon an overvoltage, some power supply designs may latch off while others do not (see the detail specification). It may be necessary to take the input voltage to zero in order to reset the output protection circuitry. The remote ON/OFF interface cannot be used to reset the output protection circuitry.

7.4.4 Hold-up time. Hold-up time refers to a power supply's ability to provide regulated output for a period of time after input power to the supply has been removed. APSP are specified to have a 50 usec hold-up time. When 270 VDC is used as the primary power to the APSP, hold-up time can be increased by adding hold-up circuitry (generally, added capacitance) across the input lines.

7.5 Status signals. APSP may be equipped with status signals which indicate various operating parameters of the supply. The power supply detail specification should be referenced to determine which status signals are included on each supply.

7.5.1 Overtemperature indication. APSP supplies are required to contain two (2) overtemperature indication interfaces (see figure 5). Both of these interfaces are thermally activated, normally open switches which close when the overtemperature condition is satisfied.

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7.5.1.1 Unsafe temperature indication. The unsafe temperature interface is nominally set to indicate a temperature of 10°C above the worst case maximum component temperature evident with an interface temperature at the TCCT level (85°C rib for family A2 and 95°C air temperature for family A1). Unsafe temperature provides indication that internal components are operating at or near their maximum thermal limit. Power supplies operated with an unsafe temperature indication are subject to damage or failure.

7.5.1.2 Excessive temperature indication. The excessive temperature interface is nominally set 11 to 15°C lower than the unsafe indication. Excessive temperature indication is given to allow attention to the unit before unsafe indication is given.

7.5.2 Power interrupt. The power interrupt signal is available from the power supply to indicate loss of input voltage and impending loss of output voltage. The power interrupt signal, hold-up time, and turn-off threshold are interrelated. See the detail specification for specific requirements for each power supply.

7.5.3 Output status. The output status function acts as a monitor for each output voltage on the unit. Status will be indicated in two ways: (1) a green LED located on the power supply cover, (2) a status signal interface at the electrical connector. The "Go" condition is indicated by a lighted LED or by a low (less than 0.4 Vdc) status signal. The status signal logic is specified such that an out of tolerance condition is indicated by an "off" light or a logic high or high impedance state. This way any power supply out of tolerance condition, including total loss of input or output power, will give a fault indication.

7.5.4 Malfunction. The malfunction signal is an indicator of internal fault condition or interruption of input power. Malfunction status is indicated in two ways: (1) a green LED located on the power supply cover, and (2) a status signal interface at the electrical connector. Normal operation is indicated by a lighted LED or by a low level (less than 0.4 VDC) malfunction signal. High level signal or LED off state occurs when one of the conditions in table I is encountered. It should be noted that interruption of input power will prompt the LED off and may not be indicative of internal, power supply related, malfunction. The malfunction signal is intended to allow fault isolation to the faulty power supply in a paralleled configuration.

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8. ELECTROMAGNETIC INTERFERENCE

8.1 Introduction. Switching-mode power supplies inherently generate electrical noise due to input AC rectification and high-frequency switching circuitry. As a result, consideration must be given to minimizing interaction of this noise in the form of electromagnetic interference (EMI) with other electronic equipment. This minimization can only be achieved through a joint effort in both power supply design and system level solutions.

8.2 Low frequency conducted emissions (CE01). Power is processed in SPS in two stages: input AC rectification and output DC regulation. The source of low frequency (60 Hz to 15 kHz) conducted emissions (CE01) is the rectification of the AC input power. The switching action of the power supply input rectifier distorts the normally sinusoidal input current waveform and establishes harmonic current flow on the AC line.

8.2.1 CE01 limits. With three-phase rectification and filtering, the input current harmonic content for SPS exceeds the CE01 requirements of MIL-STD-461. As a result, modified CE01 limits are imposed on SPS. For SPSP power supplies, this limit is based on the requirements of MIL-STD-1399, section 300. Relaxations are allowed in the 300 Hz to 1500 Hz range since MIL-STD-1399 requirements can only be met under ideal conditions as well. Relaxation is also allowed in the 4 kHz to 15 kHz range for the /1, /2, and /3 (SPSP) power supplies which are required to have external input filtering (see 8.3.1 and 8.4) for attenuation of high frequency conducted emissions (CE03).

8.2.2 Some system level CE01 solutions. Electronic solutions for complying with MIL-STD-461 CE01 limits exist but cannot practically be implemented in SPS designs. Design solutions often require penalties in size, weight, complexity, and cost of SPS -- oftentimes offsetting the benefits gained from using switching-mode technology in the first place. These considerations can be better justified at a system level where more than one power supply may benefit from a single point source solution.

8.2.2.1 DC input power. When DC input is utilized, there are no powerline harmonic currents generated by the power supply so compliance to MIL-STD-461 may be met. NAVMAT P-4855-1A states that the advantages of using DC power should be considered when selecting prime power for electronic systems. The Navy Standard Electronic Power System, as described in MIL-STD-2036, is one method for providing 155 VDC input as prime power for SPS.

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8.2.2.2 Multi-phase transformers. Another approach involves the use of multi-phase transformers which reduce low frequency harmonic currents by increasing the number of phases to be rectified. Once rectified, the DC output of the multi-phase transformer can be provided as input to SPS. Multi-phase transformers designed specifically for use with SPS are now commercially available.

8.2.2.3 Line filtering. A third approach involves passive line (LC) filtering. Consideration must be given to the fact that LC filters used for CE01 attenuation may be large and require "tuning" for specific current components.

8.3 High frequency conducted emissions (CE03). High frequency (15 kHz to 50 MHz) conducted emissions are created by switching devices in the DC-DC converter block of the SPS. Power transistors and diodes switch high voltages quickly--producing harmonics of the switching frequency back onto the input lines.

8.3.1 CE03 limits. Depending on power supply type, the supply may or may not be specified to meet MIL-STD-461 CE03 requirements. Filtering external to the powre supply may be required to meet the system level MIL-STD-461 requirements. For example, the SPSP /1, /2, and /3 power supplies were originally developed for use in the Navy's AN/BSY-1 weapons system. As a result of the requirement for power supplies with high power density and due to the technology available during the design phase of these supplies, CE03 filtering requirements were realized by external line filters which could be used by more than one power supply. Subsequently, these power supplies were specified to a relaxed CE03 limit--based on MIL-STD-461 with a relaxation equal to an adequate line filter attenuation level shown on figure 6.

8.4 Filter design. Some special considerations must be made in the design of filters to be interfaced with switching-mode power supplies.

8.4.1 Stability. Switching-mode power converters are constant power devices. Incremental increases in input voltage levels cause proportional incremental decreases in input current. That is, SPS exhibit negative input resistance characteristics. The negative input impedance of the power supply in combination with the output impedance of the filter can, under certain conditions, create a negative impedance oscillator--causing input instability.

8.4.1.1 Impedance analysis. Stability can be maintained by ensuring that source impedances are less than the power supply input impedance at the power distribution/power supply interface

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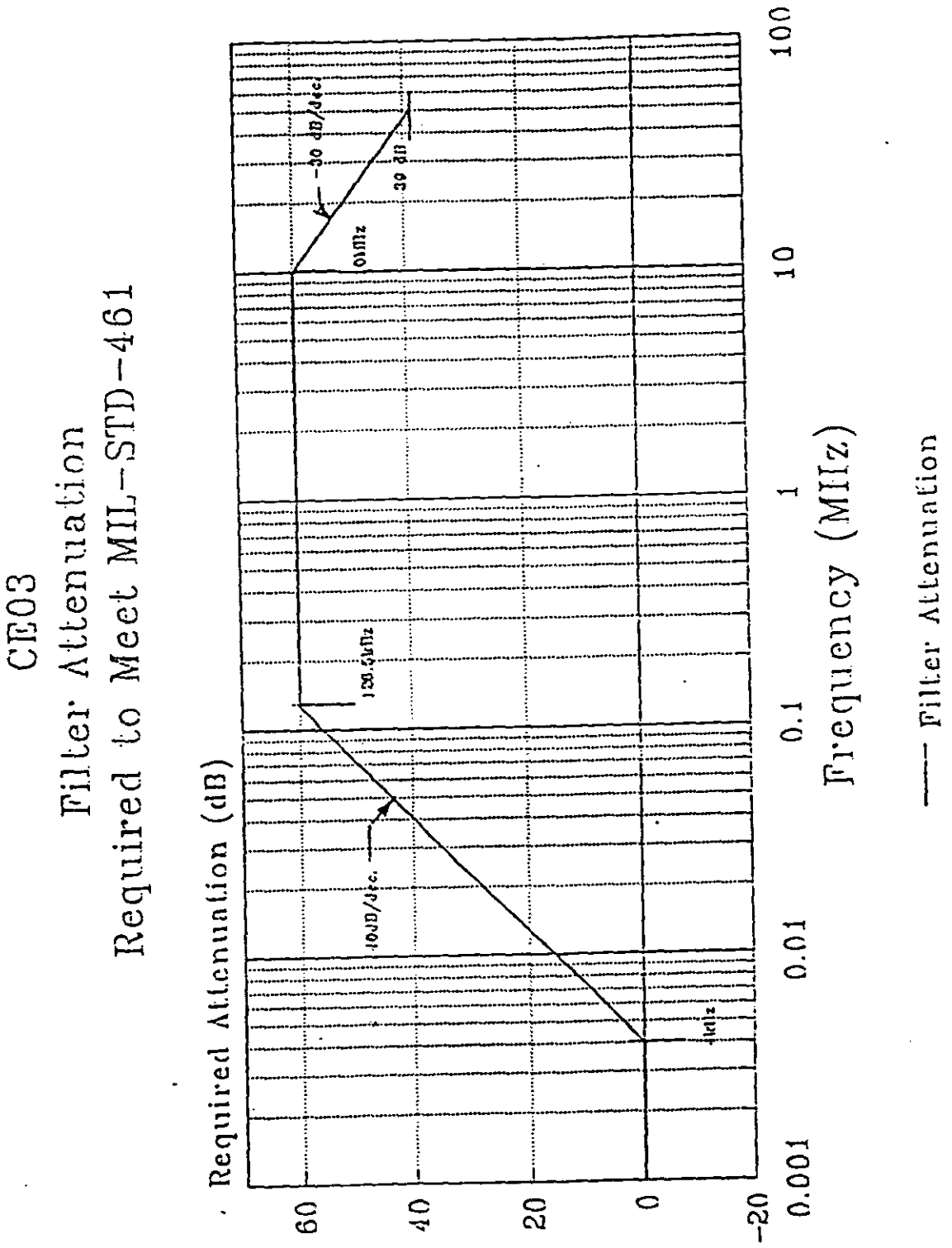


FIGURE 6. CE03 filter attenuation required to meet MIL-STD-461.

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over a frequency range from the primary power frequency up to and including the filter crossover frequency. Various power distribution components such as cabling impedance, source impedance, as well as the EMI filter impedance must be considered as sources of interaction with the power supply input impedance. Stability analysis should be investigated under worst case operating conditions to ensure acceptable system performance. Worst case conditions include the minimum power supply input voltage including droop or brownout conditions. MIL-HBK-241 provides guidelines for designing EMI filters to accommodate stable systems.

8.4.1.2 Impedance curves. SPS detail specifications contain curves of the minimum input impedance allowed on each supply. In addition, each manufacturer of SPSP is required to generate computer models of the supply input impedance for use by system designers in stability analysis. These models may be obtained from the SPSP-QAA or the APSP-QAA.

8.4.2 Power rating. EMI filters should be designed to attenuate conducted emissions at the input power level at which the power supply will be operating. Overdesign (allowing for higher input levels) may lead to inadequate attenuation at the actual operating condition.

8.4.3 Damping at resonant frequencies. Adequate damping should be included to minimize resonant peaking in input filters. Excessive peaking may cause the source inpedance to exceed the power supply input impedance causing instability.

8.5 Radiated magnetic field emissions (RE01). Magnetic field emissions are created when current loops flow in a circuit. In SPS, resulting magnetic fields are caused by both CE01 and CE03 current flow.

8.5.1 RE01 limits. SPS are specified to meet the RE01 requirements of MIL-STD-461C with the measuring device placed at a distance of 7 centimeters (cm) from the surface of the supply. MIL-STD-461 requirements are specified to an upper frequency of 50 kHz. Since SPS may regulate at frequencies in excess of 200 kHz, additional limits are specified from 50 kHz up to 10 MHz to control emissions generated by switching components. Refer to the detail specification for specific RE01 limits.

8.5.2 Limiting RE01 emissions. Magnetic field emissions can be minimized at a system level by reducing loop paths for current to flow. Crosstalk between interconnecting wires and cables can be minimized by one or more of the following.

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(a) group like cables into bundles but do not mix cables from different groups (that is, ac power cables, dc power cables, low level analog control lines, etc.).

(b) route ac power cables along the bottom and sides of the cabinet chassis.

(c) treat dc power cables the same as ac power cables but separate dc and ac cables as far as possible from each other.

(d) for analog control lines, use shielded cabling and route along bottom or sides of the chassis using the shortest run possible.

(e) do not suspend a cable bundle above the chassis. This will minimize shielding provided by the chassis.

(f) minimize the loop area formed by interconnecting wires or cables.

(g) use twisted pairs of wires where possible.

8.6 Radiated electric field emissions (RE02). Electric field emissions are caused by "instantaneous" directional changes in current flow. Current pulses caused by the switching action of the output regulator create electric field emissions.

8.6.1 RE02 limits. Family S1 SPSP are intended for use in Navy CEEE cabinet enclosures. When configured this way, power supplies are mounted on the outside of the cabinet drawers with the connector opening adjacent to interior circuitry (see figure 2). The case of the power supply is external to the cabinet. As a result, family S1 SPSP are required to meet two distinct RE02 requirements: installed and connector opening configurations.

8.6.1.1 RE02 installed configuration. The installed configuration takes into account power supply radiated emissions through the case to the environment external to the cabinet. These emissions include what may be radiated to other nearby cabinets and systems. RE02 installed configuration must meet MIL-STD-461C requirements since radiated emissions through the power supply case to the outer environment must meet the same requirement that the cabinet must meet.

8.6.1.2 RE02 connector opening configuration. The connector opening configuration considers emissions from the connector interface which circuitry internal to the cabinet can pick up. If the cabinet is a sufficient shield, emissions radiated from the connector opening will not escape the cabinet

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to the outer environment (also, emissions from the power supply case will not penetrate the enclosure and affect internal circuitry).

8.6.2 Shielding. In cases where SPS are to be used in cabinets or enclosures other than the CEEE, shielding consideration for radiated emissions must be made. Some of these are:

(a) when the enclosure has removable panels, drawers, etc., use EMI gasketing material around the openings. Gasket integrity should be checked at all points along the opening to ensure a good electrical contact.

(b) use overlapping, tight seams in corners and joints in the construction of the enclosure. Gaps in seams and joints provide areas for radiation or penetration of radiated EMI.

(c) avoid metal to metal contact of painted surfaces. If paint is necessary for protection against corrosion or oxidation, then use conductive paint.

(d) for enclosures with hinged covers, attach flat bonding straps at the hinged point between the enclosure and cover. The straps should be kept as short as possible. EMI gasketing material should be used around the remaining periphery.

(e) ventilation openings should be covered with perforated grids or slots. The size of the grid openings or slots should be as small as possible.

8.6.3 Bonding. SPS may have 4 chassis contacts at the connector which need to be connected to the cabinet chassis. In order to provide a good radio frequency (RF) current return path, the bond must be low in impedance. In order to present a low impedance path, the type of connection, type of bonding material, and attention to corrosion prevention are important factors.

8.6.3.1 Connections. The best connection (for bonding straps, wires, etc.) is a permanent connection such as welding or brazing. Sheet metal, or other screw types are not preferred for bonding. Both surfaces should be clean and free of foreign material including anodizing and adhesives. Following the bonding, the connection must be sealed to prohibit the ingress of moisture, which can be a catalyst for the corrosion process.

8.6.3.2 Bonding materials. The best grounding materials are solid, flat metallic conductors. Woven braid is not as good

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as a flat strap because oxidation can occur between the strands. Broken strands in metallic conductors can be a source or receptor of EMI.

8.6.3.3 Corrosion prevention. Corrosion can be caused by metal to metal contact by two different methods: galvanic potential difference of two different metals and local currents flowing through either two different or two like metals. Galvanic corrosion is caused by contact of two dissimilar metals and requires an electrolyte (moisture acts as an electrolyte). The greater the difference of the galvanic potentials of the two metals, the greater the potential of corrosion. The bonding materials should be low in electrochemical action, such as copper or nickel, or the dissimilar metals should be near in the electrochemical series.

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9. DISTRIBUTION

9.1 Introduction. Very seldom does an electronic system require just one power supply to drive the system load. Generally, more than one supply is used to provide distributed power throughout the system, to increase reliability through redundancy, or for both reasons. This section provides information on using SPS in distributed architectures and some considerations on reliability benefits through redundancy.

9.2 Distribution. Distribution of power converters can be categorized into two (2) distinct architectures. Congregated architectures are configured such that power supplies, connected to a common output bus, power the entire load group. Family S1 SPSP and family A1 APSP are particularly suited for this application. Dispersed architectures place one or more power supplies near individual load groups. Family S2 SPSP and family A2 APSP, which are packaged as SEM compatible units, are well suited for this application since these supplies have a form factor compatible with the loads they are supplying and can be located near these loads in the system rack.

9.3 Redundancy. There are two major classes of redundancy.

(a) Standby redundancy. Standby redundancy requires external components to detect, decide, and switch to another element. Upon detection of a failure, system operation is switched to a backup or standby unit.

(b) Active redundancy. Active redundancy does not require external components for detection, decision, and switching when an element or path in the system fails.

9.4 Mean time between failures. Mean time between failures (MTBF) is a concept used to indicate the probability of successful operation of a piece of equipment over a mission life based on the inherent reliability of the equipment design. MIL-HDBK-217 establishes criteria for the prediction of MTBF based on component stresses and operating temperatures. SPS are designed for a minimum MTBF of 100,000 hours while operating under full-load conditions at a baseplate temperature of 60°C.

9.4.1 Failure rate. Assuming that the rate of failure for a piece of equipment is constant over its mission life, then the failure rate (λ) is defined as:

$$\lambda = 1/\text{MTBF} \qquad \text{(eqn. 9-1)}$$

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Having a MTBF equal to 100,000 hours does not mean that each power supply will last 100,000 hours before it fails. Instead, the failure rate during any one hour is 1/100,000 or 0.001 percent. Given the assumption of constant failure rate, the probability (R(t)) of a power supply not failing prior to some time (t) is:

$$R(t) = e^{-\lambda t} = e^{-t/MTBF} \qquad \text{(eqn. 9-2)}$$

The probability for not failing exponentially decays over the mission life. It should be noted that the probability of successful power supply operation for t = MTBF = 100,000 hours is a non-zero value.

$$R(t) = e^{-100,000/100,000} = 36.8 \text{ percent}$$

9.5 Standby redundancy with SPS. Standby redundancy can be implemented with SPS using an ON/OFF paralleling scheme. One power supply is operated in the ON state while the second supply is kept in a standby or OFF state ready to be switched ON in the event of failure in the first supply (figure 7). Sensing and control circuitry is needed to detect an output failure in the operating unit and to prompt the standby unit into an ON condition. This type of paralleling can be implemented using the remote ON/OFF function of the power supply when this function is available.

9.5.1 Total load capacitance. In the ON/OFF paralleled configuration, the loading on the energized power supply is a combination of the actual load and the output capacitance of the unenergized power supply. The user should ensure that the maximum load capacitance requirements of the power supply detail specification are not exceeded for this configuration.

9.5.2 Pre-bias considerations. Paralleled supplies prompted on by toggling the input power (instead of the remote ON/OFF interface) while retaining a residual voltage present from the failed supply at the paralleled output may sense an overvoltage condition at startup and latch off themselves. This problem can be overcome by using the remote ON/OFF interface only for ON/OFF paralleling control.

9.5.3 Switching delay. Some switching time delay will be evident when a failure occurs in the ON/OFF paralleled configuration. This delay is a combination of the inherent time delay of the sense/control circuitry and the remote turn-on delay for the standby power supply (SPSP are specified to have less than a 250 msec remote turn-on time). Steps should be taken to

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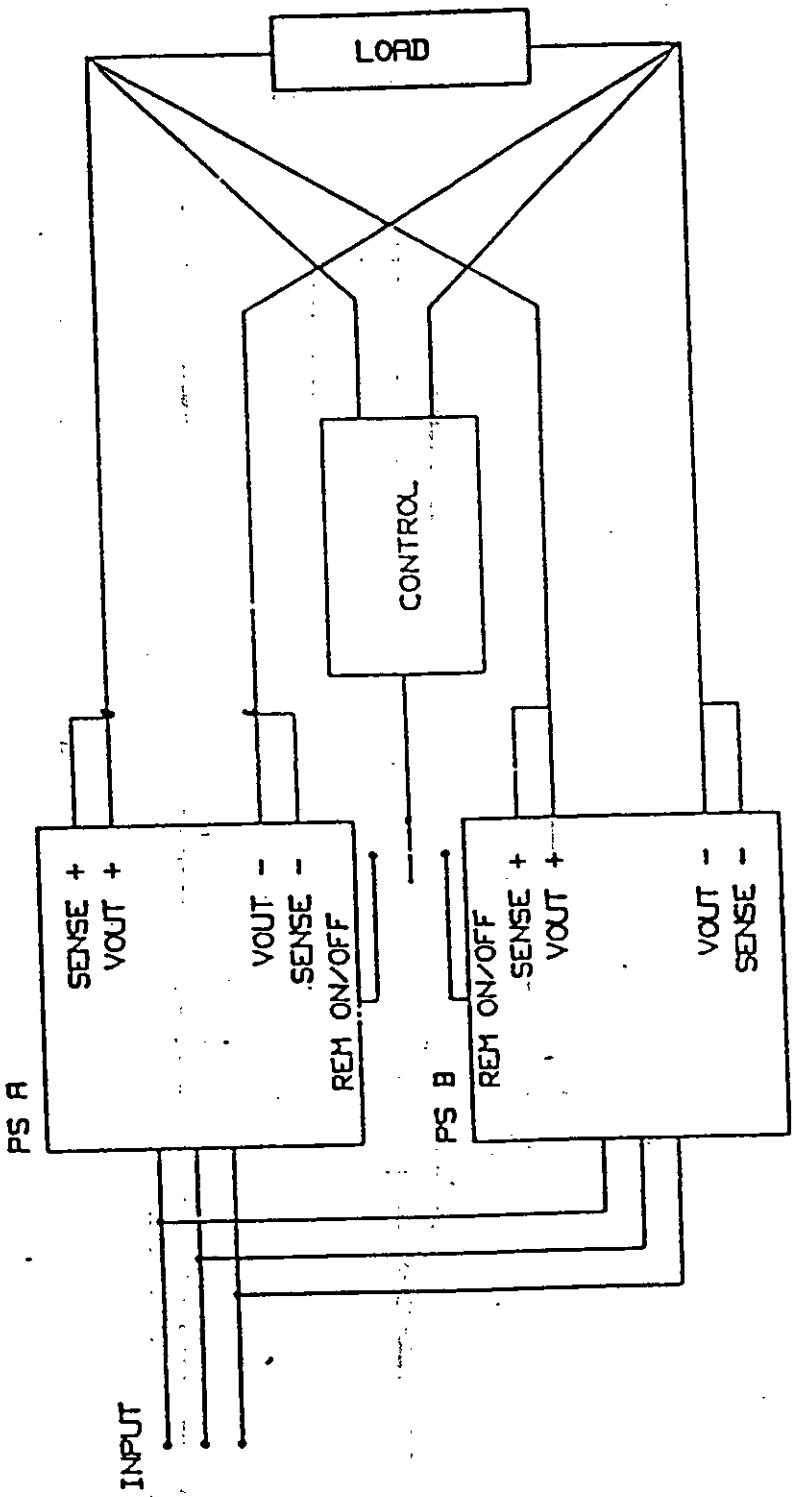


FIGURE 7. Standby redundancy configuration.

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ensure that mission essential loads can overcome the effects of associated voltage excursions.

9.5.4 Output status signal. Attempts have been made to use the output status signal for controlling power supplies in the ON/OFF configuration. The power supply specification allows up to a maximum of 500 mV peak of noise on this status signal. Generally, circuitry needed to use this signal is located at some distance from the power supply in the system rack and long leads are used as an interface. Additional noise coupled on these leads combined with the noise already evident may present a problem when using Fast TTL or CMOS logic devices to sense the status signal. System use in this manner without attention to cable routing, termination, or application may lead to inadvertent or false shutdown/startup of the paralleled supplies. The system designer should be aware that some form of signal conditioning or EMI control may be mandated when using the output status signal in this manner.

9.6 Active redundancy. Active redundancy can be implemented with some SPS using the diode ORed parallel configuration (figure 8). OR-ing diodes are needed to isolate the two outputs from each other, decreasing the possibility of a single point of failure in the event of output short circuit of either supply. Sense lead location is a factor in performance of power supplies configured diode-ORed. Generally, the sensing point should be made on the load side of the diodes as illustrated. However, on some supplies, which have a tighter tolerance on the output regulation, some current sharing between supplies may be obtained by placing the sense leads on the power supply side of the OR-ing diodes.

9.6.1 Current sharing. SPS configured in a diode ORed configuration will probably not current share very well. The output voltage of two paralleled power supplies will not be exactly equal. Subsequently, the supply at the higher voltage will try to supply all of the load current (up to the power supply's output current limit). In applications where the current required by the load is significantly less than the individual power supply current limit, the power supply at the lower output voltage level will effectively be forced into an OFF state (that is, it will supply none of the load current). In certain applications, this effect is desired. One supply can be intentionally programmed to a higher output voltage (such as 5.25 V versus 5.05 V) such that it forces the second supply into the OFF condition -- effectively configuring the ON/OFF configuration without a need for the extra sense/control circuitry.

9.6.2 Output status indication. The output status signal may or may not indicate its own condition when other power

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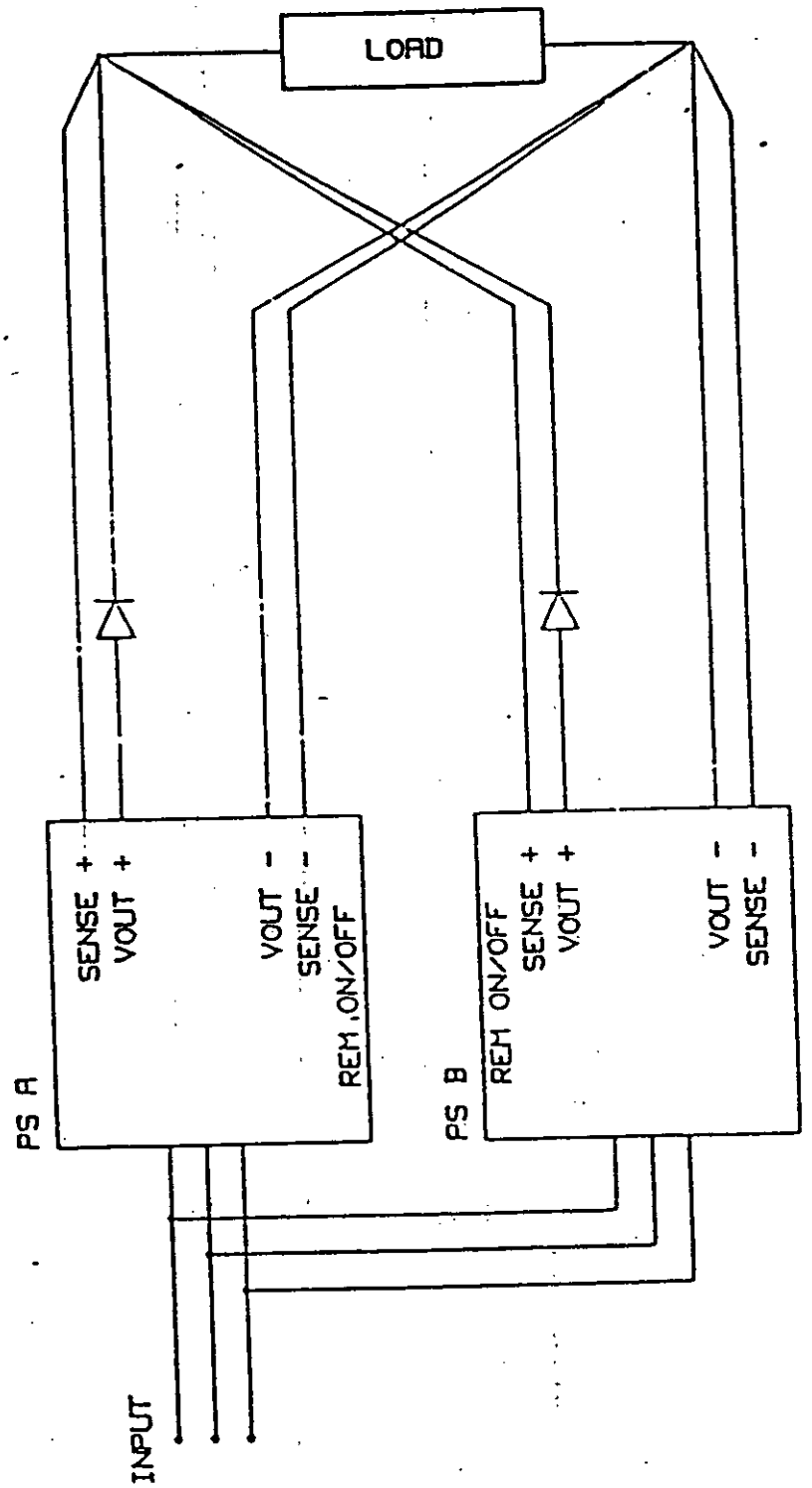


FIGURE 8. Active redundancy configuration.

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supplies are paralleled with it in the ON/ON configuration. Even using OR-ing diodes, false "good" indication may be provided by the status signal. Some SPS designs monitor the output condition from the output sense lead while others monitor the output itself. An example of false indication can be seen by referring to figure 8. Assume that power supply A fails and senses its output at the sense lead location. With power supply B operational, unit A will still sense the proper output at the sense lead location and indicate a false "good" status.

9.6.3 Reliability for paralleled SPS. The effective MTBF for SPS paralleled in the ON/ON configuration on figure 8 where N units are required to supply the load is given as:

$$MTBF_{eff} = \frac{MTBF}{N} \qquad \text{(eqn. 9-3)}$$

(N = the number of required units)

EXAMPLE

Two (2) AA3A power supplies which can supply 150 A each are paralleled in an ON/ON configuration to drive a 300 A load. Each supply has a MTBF = 100,000 hours. The system $MTBF_{eff}$ is:

$$MTBF_{eff} = \frac{100,000}{2} = 50,000 \text{ hours}$$

9.6.4 N+1 redundancy. It is possible to parallel an additional power supply to those already needed to power the system and gain benefits in overall reliability. If N+1 power supplies are paralleled in an application where only N power supplies are required, the effective MTBF of the system becomes:

$$MTBF_{eff} = \frac{MTBF}{(N+1)} + \frac{MTBF}{N} \qquad \text{(eqn. 9-4)}$$

EXAMPLE

Three (3) AA3A power supplies which can supply 150 A each are paralleled in an ON/ON configuration to drive a 300 A load. Each supply has a MTBF = 100,000 hours. The system $MTBF_{eff}$ is:

- only 2 supplies are actually needed to power the load so N = 2

$$MTBF_{eff} = \frac{100,000}{(2+1)} + \frac{100,000}{2} = 83,333 \text{ hours}$$

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9.6.4.1 Redundancy for N=1. For systems which can be powered by one (1) power supply, the effective system MTBF can be incrementally increased to a value greater than the original MTBF by paralleling additional units. The mean time to system failure for a multi-component system in which only one (1) unit is actually needed to power the load is given as:

$$MTBF_{eff} = (MTBF) \times \sum_{n=1 \text{ to } N} (1/n) \qquad \text{(eqn. 9-5)}$$

(N = the total number of paralleled units)

Table II shows the additional increase in MTBF for each additional parallel element.

TABLE II. MTBF for incremental redundant elements.

n	MTBFeff	Δ MTBF
1	1	0.00
2	1.5	0.50
3	1.833	0.33
4	2.083	0.25
5	2.283	0.20
.	.	.
.	.	.
10	2.929	0.10

In general, the reliability gain for additional redundant elements decreases rapidly for additions beyond a few parallel elements. A two element redundant configuration will have a 50% increase in the MTBF over a single system. In the case of equal current sharing power supplies, the MTBF can be expected to increase both from the parallel configuration and from the fact that each supply is only stressed to approximately half the load current. Under these circumstances, the system MTBF would be more than 50 percent greater than the single element.

9.6.5 Forced current sharing. Some SPS may be equipped with communication circuitry to force nearly equal current sharing between paralleled supplies. One implementation of forced current sharing uses a signal generated by each paralleled supply. The signal level is proportional to the power supply output current. The signals from each paralleled supply are shared to exchange load current levels. The paralleled supplies adjust their individual outputs to match the shared rated load. The detail specification should be referenced for the characteristics of the signal.

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APPENDIX

FAMILY S1 POWER SUPPLY DIMENSIONS AND KEYING

10. SCOPE

10.1 Scope. This appendix contains the case dimensional requirements, keying locations, and key coding for family S1, Shipboard Power Supply Products (SPSP). This appendix is a mandatory part of the handbook. The information contained herein is intended for guidance only.

20. APPLICABLE DOCUMENTS

20.1 Non-Government publications. The following document forms a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DoDISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DoDISS are the issues of the documents cited in the solicitation.

AMERICAN NATIONAL STANDARDS INSTITUTE (ANSI)

ANSI Y14.5M - Dimensioning and Tolerancing. (DoD adopted)

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018-3308).

30. GENERAL REQUIREMENTS

30.1 Case styles. Family S1 power supply case styles conform to the dimensions shown on figure 16 and in table III.

30.1.1 Hold down screws. The location and number of hold-down jackscrews conforms to figure 9. Case style 1 power supply does not possess jackscrews in the V dimension.

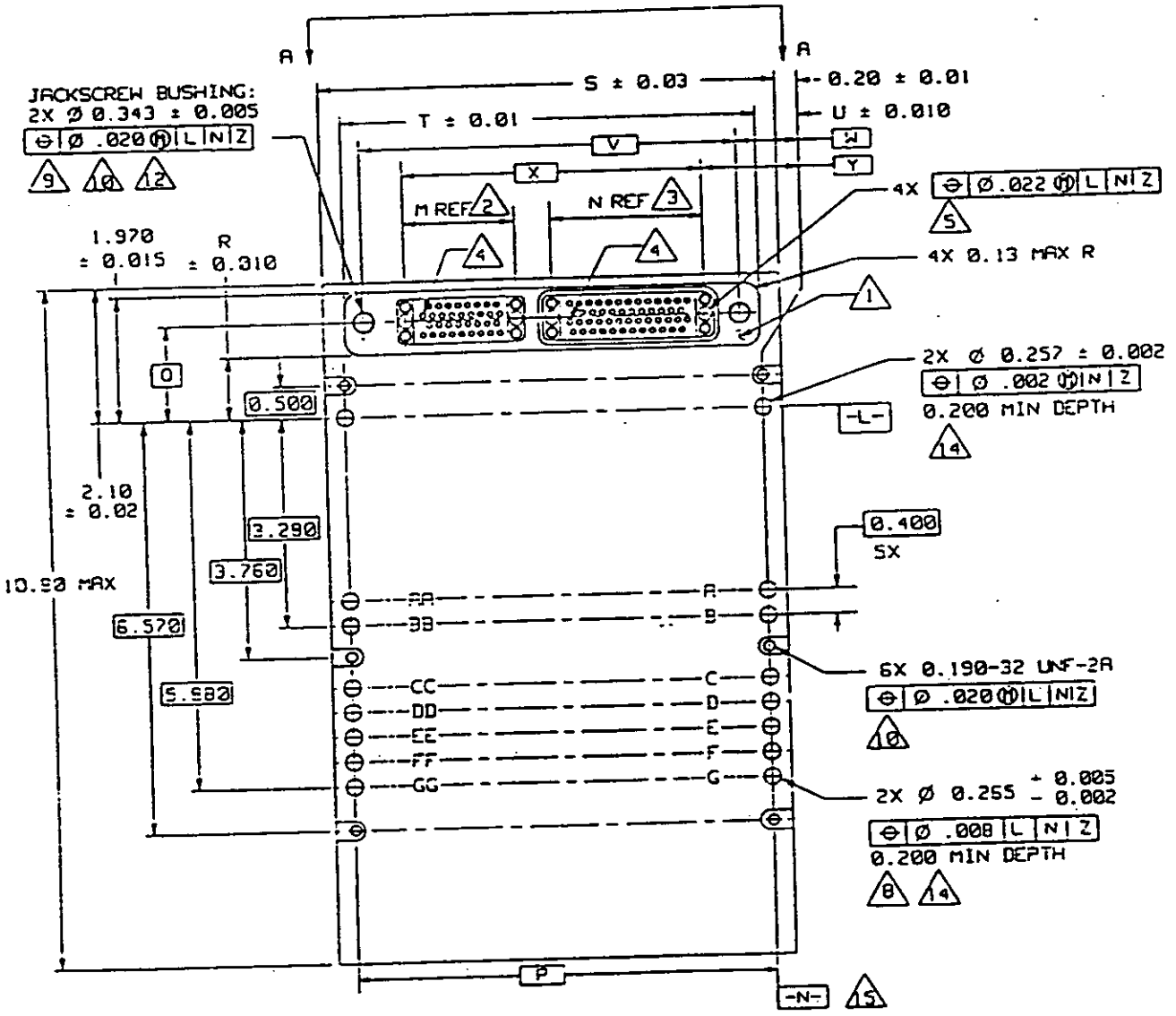
30.2 Keying. Each family S1 power supply is provided with four (4) keying holes. Two of these holes are stationary and common to all power supply styles. Two additional holes are provided as personality keying holes and are unique to each power supply type. Figure 9 shows the location of these keying holes.

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APPENDIX

30.3 Key code. The key code for each power supply type describes the location of the keying holes on the input connector side of the power supply baseplate, the case style, and the location of the keying hole on the output connector side of the power supply.

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BOTTOM VIEW

FIGURE 9. Family S1 dimensions and key code.

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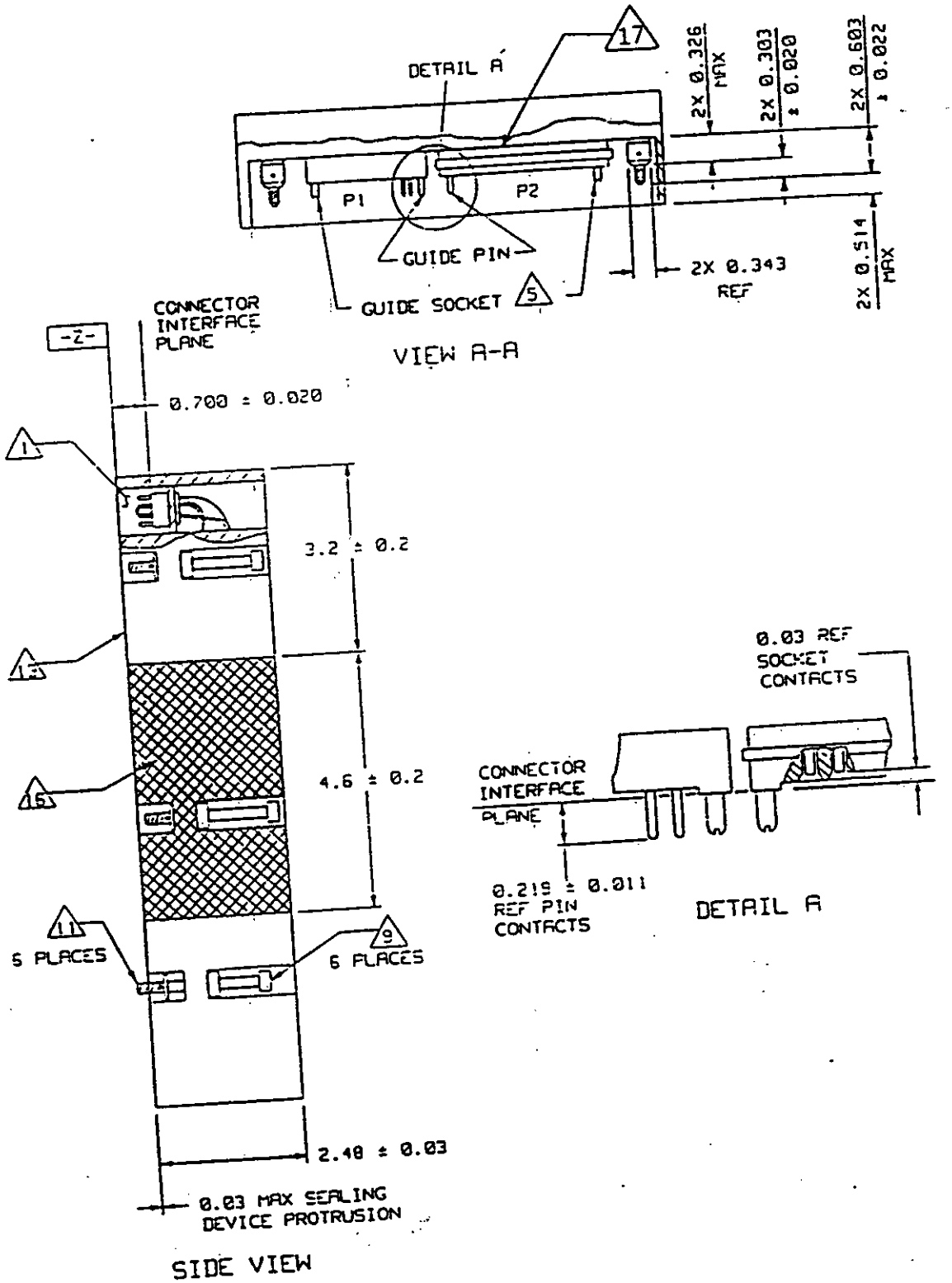


FIGURE 9. Family S1 dimensions and key code - Continued.

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NOTES:

1. Connector interface recess. Connector recess dimensions shall be held to a depth of 1.020 minimum to allow clearance for the mating connector.

2. Input connector.

3. Output connector.

4. Indicates pin "A" on connector.

5. Install guide socket in this location.

6. 0.062 minimum contact wiping engagement is required between the power supply connectors and the mating connectors for full current carrying capacity of the contacts.

7. The entire case shall provide environmental protection (including corrosion prevention) and electric and magnetic EMC shielding as prescribed in the associated detail specification.

8. View illustrates all possible personality keying hole locations. Refer to associated detail specification for actual key code and corresponding keying hole locations.

9. 0.190-32 UNF-3A socket head cap screw. Hexagonal keyway in screw head to measure 0.1562 + 0.0025 - 0.0000 across flats.

10. Dimensions and tolerances control location of mounting hole for captive screw. Captive fasteners must provide a minimum of 0.015 and a maximum of 0.04 radial float from nominal position.

11. 0.340 $\pm$ 0.030 penetration of captive fasteners in the secured position. In the retracted position, the ends of the fasteners shall recess 0.010 minimum into the mounting surface.

12. Jackscrew fasteners at the connectors are deleted on style 1 power supply.

FIGURE 9. Family S1 dimensions and key code - Continued.

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NOTES: - Continued.

13. 0.0005 inch/inch and 0.005 overall flatness and 32 microinches finish tolerances to be maintained on mounting surface.

14. 1.00 minimum pin clearance past -Z- shall be provided. The initial 0.200 of hole depth shall be as specified. The diameter of the remainder of the pin clearance hole may be enlarged, but shall not be less than the minimum diameter of the initial 0.200 of hole depth and shall be positioned to guarantee clearance around the initial hole perimeter.

15. -N- is perpendicular to -L- at the center of the 0.257 diameter hole.

16. A gripping surface shall be located in the designated area on both sides of the power supply.

17. This surface, as defined by the dimensions on the connector interface plane and connector body, is not required to be (but is not prohibited from being) the same plane as defined by the dimensions on the jackscrew and bushing.

18. Dimensions are in inches.

19. Dimensioning is in accordance with ANSI Y14.5M.

FIGURE 9. Family S1 dimensions and key code - Continued.

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TABLE III. Dimensions for figure 9.

Style	Dimensional variables											
	P	Q	R	S	T	U	V	W	X	Y	M	N
1	4.600	1.480	0.980	5.00	4.40	0.100	N/A	N/A	3.450	0.575	1.255	1.694
2	5.400	1.480	0.980	5.80	5.35	0.025	4.750	0.325	3.880	0.760	1.694	1.694
3	6.400	1.480	0.980	6.80	6.38	0.012	5.900	0.250	4.437	0.982	1.694	2.290
4	8.400	1.163	0.360	8.80	6.80	0.800	6.300	1.050	4.850	1.775	1.976	2.389
5	11.400	1.163	0.360	11.80	6.80	2.300	6.300	2.550	4.850	3.275	1.976	2.389

All dimensions are in inches.

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CONCLUDING MATERIAL

Custodians:
Army - ER
Navy - SH
Air Force - 99

Review activity:
DLA - GS

Preparing activity:
Navy - NW

(Project 6130-0349)

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

INSTRUCTIONS

1. The preparing activity must complete blocks 1, 2, 3, and 8. In block 1, both the document number and revision letter should be given.
2. The submitter of this form must complete blocks 4, 5, 6, and 7.
3. The preparing activity must provide a reply within 30 days from receipt of the form.

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I RECOMMEND A CHANGE:	1. DOCUMENT NUMBER MIL-HDBK-845	2. DOCUMENT DATE (YYMMDD) 31 December 1992

3. DOCUMENT TITLE
Standard Power Supply Applications Handbook

4. NATURE OF CHANGE *(Identify paragraph number and include proposed rewrite, if possible. Attach extra sheets as needed.)*

5. REASON FOR RECOMMENDATION

6. SUBMITTER

a. NAME (Last, First, Middle Initial)	b. ORGANIZATION	
c. ADDRESS (Include Zip Code)	d. TELEPHONE (Include Area Code) (1) Commercial (2) AUTOVON (If applicable)	7. DATE SUBMITTED (YYMMDD)

8. PREPARING ACTIVITY

a. NAME	b. TELEPHONE (Include Area Code) (1) Commercial	(2) AUTOVON
Technical Point of Contact (TPOC): Mr. Mike Roach (Code 6023) PLEASE ADDRESS ALL CORRESPONDENCE AS FOLLOWS:	TPOC: 812-854-5106	88-482-5106
c. ADDRESS (Include Zip Code) Commander Code 6023, Bldg. 2917 (Mike Roach) NAVSURFWARCENDIV, 300 Highway 361 Crane, Indiana 47522-5001	IF YOU DO NOT RECEIVE A REPLY WITHIN 45 DAYS, CONTACT: Defense Quality and Standardization Office 5203 Leesburg Pike, Suite 1403, Falls Church, VA 22041-3466 Telephone (703) 756-2340 AUTOVON 289-2340	

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